Square-Wave Voltammetry Forward Model

Prediction of the Voltammetric Response to MMP-9 Activity

Eton iGEM 2026, Dry Lab | Layers 8, 10, 11, 13, 15 | `swv_forward_model.py`

Abstract. This report presents the forward model of the square-wave voltammetric readout of the MMP-9 sensor. The model predicts the voltammogram recorded at a gold electrode carrying a methylene blue (MB) labelled peptide probe as a function of the MMP-9 concentration and the incubation time. The signal chain is formulated in seven steps: surface composition, tether geometry, single-probe electrode kinetics under a square-wave potential program, summation over the probe population, validation, readout of the cleavage fraction from the MB peak height, and cleavage kinetics with detection limit. The peak height is proportional to the fraction of intact probe, so the cleavage fraction $F_k^{\mathrm{rec}} = 1 - \Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{MB}}(0)$ is independent of probe coverage. At the working point (10 Hz, $\Gamma_{\mathrm{tot}} = 3$ pmol cm^{-2}, $A = 0.017$ cm^2) the predicted peak is 15.9 nA at -251 mV with a width of 58 mV, arising from labels a few tenths of a nanometre above the tip of the spacer. Healing (6 nM) and chronic (30 nM) MMP-9 concentrations give cleavage fractions of 14% and 52% after 30 minutes. The absolute current depends on unmeasured inputs and is uncertain by a large factor; the detection limit depends on the readout noise and the surface efficiency of cleavage, neither of which has been measured.

Scope and conventions. The model describes the MB channel of the electrode. The electrode is prepared by one-pot co-immobilisation of the MB-peptide-PEG6 probe and ferrocene-hexanethiol (Fc-hexanethiol) without mercaptohexanol (MCH); Fc-hexanethiol acts as the diluent and is not represented in the model (Section I, Part F). The report is organised into a description of the system and measurement (Section I) followed by eight parts, A to H, each treating one step of the signal chain. Each part states its objective and result, the assumptions, the governing equations with derivations and worked values, the predictions, and the supporting evidence. Values obtained from a published source or confirmed on the fabricated device are annotated *(established)*. Working assumptions not yet confirmed on the device are annotated *(assumed)*. Symbols and abbreviations are listed in the Nomenclature.

I. System Description and Measurement Principle

Application

The matrix metalloproteinase MMP-9 is present at a higher concentration in chronic (non-healing) wound fluid than in healing wound fluid; representative values are about 6 nM in healing and about 30 nM in chronic wounds [7b]. The device estimates the MMP-9 concentration in a small sample of wound fluid from an electrical measurement on a disposable gold electrode. The model relates the MMP-9 concentration and incubation time to the electrode readout.

Electrode and probe

The sensor is a gold electrode of geometric area 1.7 mm^2 (0.017 cm^2) [12] *(established)*. Two species are attached to the gold through thiol anchors in a single preparation step (Figure 1A):

- the **probe**: the peptide SGKIPRTLTA, which contains the MMP-9 cleavage site (the bond between residues 7 and 8), joined to the gold through a flexible spacer of six ethylene-glycol units (PEG6) and carrying the redox label **methylene blue** (MB) at its free end;
- **ferrocene-hexanethiol** (Fc-hexanethiol): a six-carbon chain terminated by a ferrocene group, co-deposited with the probe in one pot without mercaptohexanol (MCH). It occupies the gold between probes and acts as the diluent that keeps probes separated [12] *(established)*.

The MB label exchanges electrons with the gold, and this exchange is the measured signal. The model describes the MB channel; Fc-hexanethiol is not represented, and Part F treats the consequences of this scope.

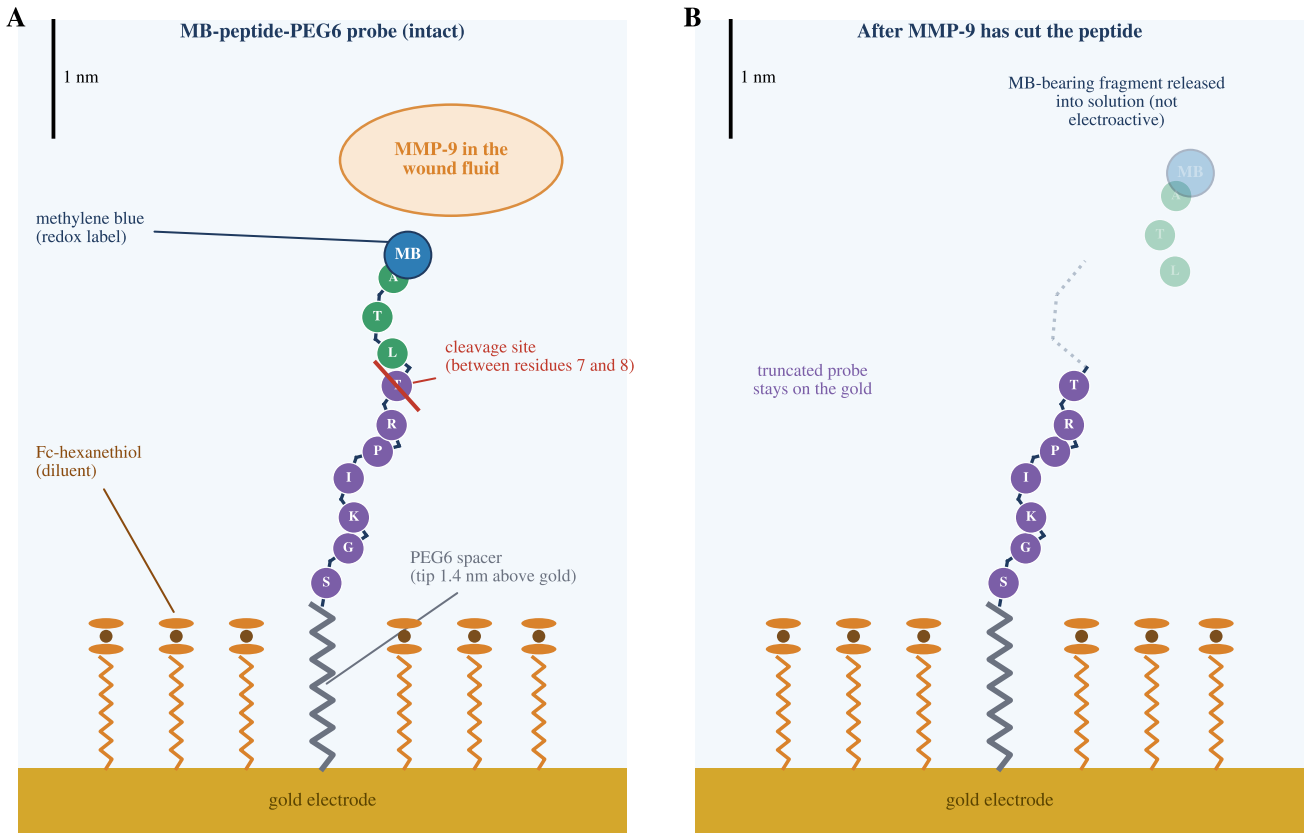


Figure 1: **A:** the electrode surface, drawn to scale in height (1 nm bar). One MB-peptide-PEG6 probe rises from the gold; the peptide is drawn in one of its many possible flexible conformations, with residues labelled S-G-K-I-P-R-T-L-T-A and the cleavage site marked in red. The shorter ferrocene-hexanethiol chains fill the gold around it. **B:** after MMP-9 cuts the peptide, the fragment carrying MB is released into solution; the stub that remains on the gold is not electroactive. Probes are on average 7 nm apart (Part A), so only one is drawn.

Measured quantity

Voltammogram. The current is a flow of electrons between the gold and the MB label. MB is either reduced (two electrons gained) or oxidised, depending on the voltage applied to the electrode relative to an Ag/AgCl reference electrode. The **formal potential** $E^{0\prime}$ is the voltage at which the two forms are equally favoured; for MB it is close to -0.25 V [1]. As the applied voltage is varied through $E^{0\prime}$ a current flows, and the graph of current against applied voltage, the **voltammogram**, has a peak near $E^{0\prime}$ (Figure 2A). The peak is characterised by its *height*, proportional to the number of MB labels participating, its *position*, near $E^{0\prime}$, and its *width*, the full width at half the maximum height (FWHM).

Effect of cleavage. Only MB labels that remain tethered to the gold exchange electrons at a rate sufficient to contribute. Cleavage by MMP-9 releases the MB-bearing fragment, so the peak height decreases in proportion to the fraction of intact probes (Figure 2C). The **cleavage fraction** F_k is the fraction of probes cleaved during an incubation.

Square-wave voltammetry (SWV). The applied voltage is stepped along a staircase and a square wave of amplitude $E_{\text{sw}} = 25$ mV is superimposed, so that at each step the voltage alternates between a forward (higher) and a reverse (lower) pulse. The current is sampled at the end of each pulse and the net current $\Delta I = I_{\text{fwd}} - I_{\text{rev}}$ is plotted against the staircase voltage (Figure 2B). The square-wave frequency is 10 to 20 Hz [12] *(established)*. The rationale for the technique is given in Part C.

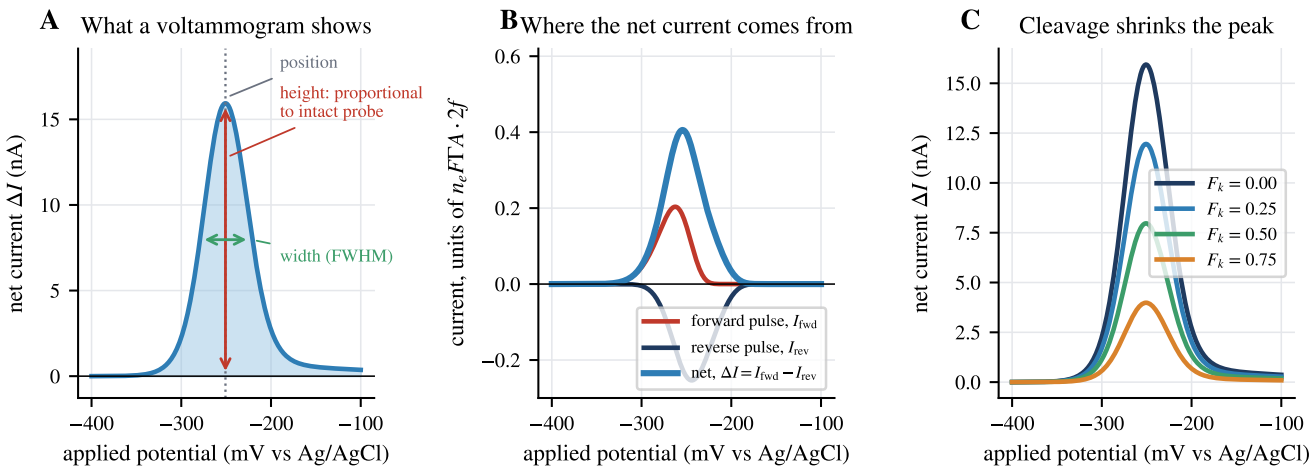


Figure 2: **A:** the model's predicted voltammogram for an intact electrode (10 Hz), with the three features that matter: height, position and width. The height is about 15.9 nA, the position −251 mV and the width 58 mV. **B:** how the net current is built from the forward-pulse and reverse-pulse currents for one probe, in units defined in Part C. **C:** as the cleavage fraction F_k grows, the peak shrinks in proportion and does not move or change shape.

Measurement protocol and readout

Each electrode is scanned twice: once *before* incubation and once *after* the sample has been on it for a time T (Figure 3). The cleavage fraction is read from the pair of peak heights on the same electrode:

$$F_k^{\text{rec}} = 1 - \frac{\Delta I_{\text{MB}}(T)}{\Delta I_{\text{MB}}(0)}, \tag{I.1}$$

where ΔI_{MB} is the net peak height. Eq. (I.1) follows because the peak height is proportional to the amount of intact probe (Part D): $\Delta I_{\text{MB}}(T)/\Delta I_{\text{MB}}(0) = 1 - F_k$. Because each electrode is compared only with its own initial scan, the number of probes it carries cancels out of F_k^{rec} (Part F). The MMP-9 concentration is then obtained from F_k^{rec} using the cleavage kinetics of Part G.

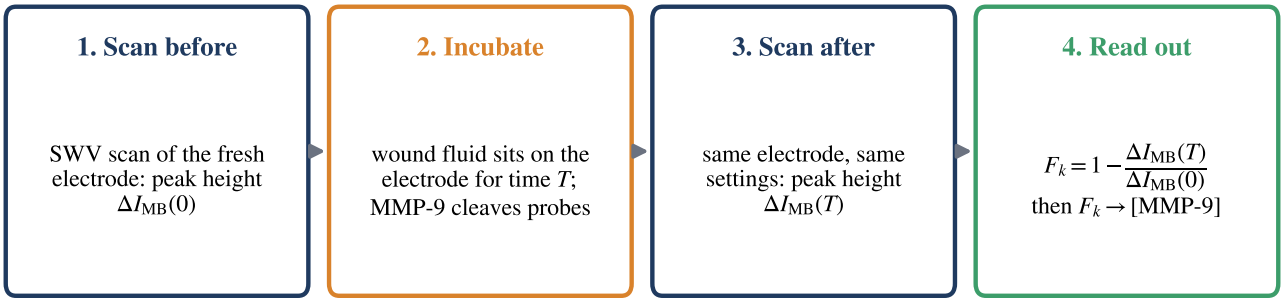


Figure 3: The four steps of a measurement. The readout uses only the two peak heights measured on the same electrode.

Structure of the model

The model follows the physical chain in Figure 4; each part of this report treats one link.

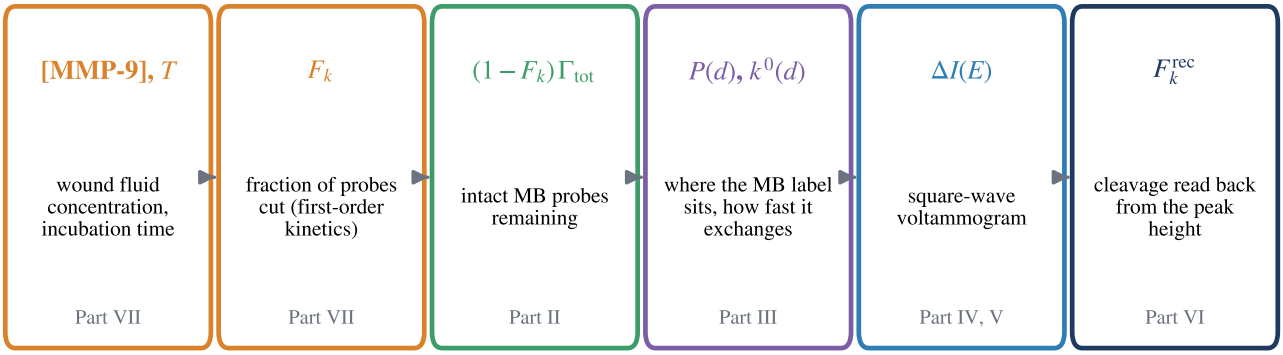


Figure 4: The forward chain from concentration to voltammogram, and the readout that closes it. The part of this report that treats each link is shown under it.

Part	Subject	Section
A	Probe coverage and spacing	Surface (II)
B	Distribution of the MB label distance and electron-transfer rate	Tether geometry (III)
C	Square-wave voltammetry; current of a single probe	Single probe (IV)
D	Electrode signal and its dependence on cleavage	Electrode signal (V)
E	Verification and validation	Validation (VI)
F	Readout of the cleavage fraction from the MB signal	Readout (VII)
G	Cleavage kinetics and detection limit	Kinetics (VIII)
H	Sensitivity to the assumed inputs	Sensitivity (IX)

Established and assumed quantities

Each quantity used in the model is either measured or published (established) or a working assumption not yet checked on the fabricated device (assumed). The principal entries are listed with the part that uses them; the complete list and the open items are given in the Summary.

Quantity	Value and source	Used in	Status
Electrode area	1.7 mm^2 [12]	C, D	*(established)*
SWV frequency, amplitude	10 to 20 Hz, 25 mV [6,12]	C, D	*(established)*
One-pot deposition of probe and Fc-hexanethiol	no MCH [12]	A	*(established)*
Solution cleavage rate, k_{cat}/K_m	67 500 $\mathrm{M}^{-1}\mathrm{s}^{-1}$ (free peptide) [7]	G	*(established)*
MB transfer coefficient, electrons	$\alpha = 0.37$, $n_e = 2$ [1,6]	C	*(established)*
Probe coverage Γ_{tot}	3 $\mathrm{pmol\,cm}^{-2}$, engineering estimate	A, D	*(assumed)*
Tunnelling decay constant β	10 nm^{-1}, from alkanethiols [4]	B	*(assumed)*
PEG6 tip height d_{PEG}	1.40 nm, Flory estimate	B	*(assumed)*
Surface efficiency η	0.20, central estimate	G	*(assumed)*
Readout noise	not measured	G	*(assumed)*
Cleavage independent of label height	assumed	F	*(assumed)*
Non-enzymatic probe loss during incubation	assumed zero	F	*(assumed)*

II. Part A: The Electrode Surface

Overview

Objective. To determine the probe density and whether probes may be treated as independent. **Result.** At the working coverage of 3 $\mathrm{pmol\,cm}^{-2}$ the electrode carries about 51 fmol of probe (3.07×10^{10} molecules), one per 55 nm^2 of gold. The mean spacing is 7.4 nm, above the distance (2.8 nm) below which the labels of neighbouring probes could overlap, so probes are treated as independent. The coverage is an assumed value; it sets the absolute

size of the current but does not enter the readout (Part F).

Assumptions

- **One-pot deposition of MB-peptide-PEG6 and Fc-hexanethiol, without MCH.** This is the protocol in use [12] *(established).*
- **Total probe coverage $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}.** An engineering estimate within the range reported for a comparable peptide-PEG system [8], not a measurement on these electrodes *(assumed).* Direct measurement is not possible with the available instrumentation [12], and the coverage is not yet reproducible from electrode to electrode.
- **The surface composition differs from the solution composition.** The peptide-PEG probe is expected to adsorb more strongly than the short Fc-hexanethiol, so the surface fraction of probe is not the fraction in the deposition solution *(assumed).*
- **Fc-hexanethiol fills the gold between probes.** Whether it forms a fully packed layer has not been determined *(assumed).* If it does, little bare gold remains and non-specific adsorption of protein onto gold is reduced, although adsorption on the layer itself is not excluded.

Equations

Probes per area and spacing. A coverage Γ_{tot} in mol cm^{-2} corresponds to $\Gamma_{\text{tot}} N_A$ molecules per cm^2, where $N_A = 6.022 \times 10^{23}$ mol^{-1} is Avogadro's number. The area per probe is the reciprocal, and for probes on a square grid the centre-to-centre spacing is its square root:

$$a = \frac{1}{\Gamma_{\text{tot}} N_A}, \qquad s = \sqrt{a}. \tag{II.1}$$

Worked example. $\Gamma_{\text{tot}} N_A = 3 \times 10^{-12} \times 6.022 \times 10^{23} = 1.81 \times 10^{12}$ cm^{-2} $= 1.81 \times 10^{-2}$ nm^{-2} (using 1 cm^2 $= 10^{14}$ nm^2), so $a = 55$ nm^2 and $s = 7.4$ nm. For randomly placed probes the mean distance to the nearest neighbour is smaller. With number density $\rho = 1/a$, the probability that a disc of radius r contains no other probe is $e^{-\rho\pi r^2}$, so the nearest neighbour lies between r and $r + \mathrm{d}r$ with probability density $p(r) = 2\pi\rho r\, e^{-\rho\pi r^2}$ (the probability that the disc is empty, times the probability that a probe lies in the ring of area $2\pi r\,\mathrm{d}r$). The mean is

$$\langle r \rangle = \int_0^\infty r\, p(r)\,\mathrm{d}r = 2\pi\rho \int_0^\infty r^2 e^{-\pi\rho r^2}\mathrm{d}r = 2\pi\rho \cdot \frac{\sqrt{\pi}}{4(\pi\rho)^{3/2}} = \frac{1}{2\sqrt{\rho}} = 3.7 \text{ nm},$$

using $\int_0^\infty r^2 e^{-ar^2}\mathrm{d}r = \sqrt{\pi}/(4a^{3/2})$ (Eq. X.3, Appendix A). The total number of probes is $\Gamma_{\text{tot}} N_A A = 3.07 \times 10^{10}$, that is 51 fmol on the $A = 0.017$ cm^2 electrode.

Fraction of a full monolayer. A full alkanethiol monolayer on gold has one molecule per 0.2165 nm^2, that is 7.7×10^{-10} mol cm^{-2}. The probe therefore occupies only 0.4% of the sites, roughly 1 in 256; the remaining sites carry Fc-hexanethiol if the layer is fully packed.

Independence of probes. The label at the end of a tether is displaced sideways from its anchor by a random amount with root-mean-square value $\sqrt{2}\,\sigma_z = 1.4$ nm (derived in Part B). Two labels can interfere only if their anchors are closer than about twice this, 2.8 nm. The spacing of 7.4 nm exceeds this (Figure 5), which justifies computing the signal of one probe and multiplying by the number of probes.

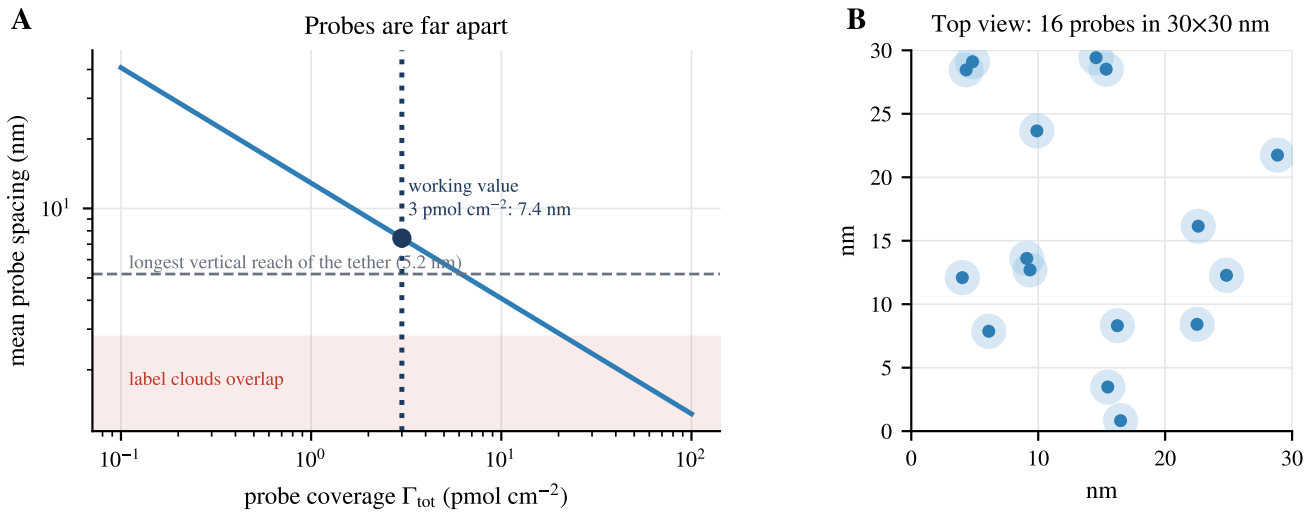


Figure 5: **A:** mean probe spacing (Eq. II.1) against coverage. Below the shaded band the sideways reach of neighbouring labels overlaps. The working value (dot) lies above it, and remains above it up to about 22 pmol cm^{-2}. **B:** a top view of a random arrangement of probes at the working coverage, with the sideways reach of each label shaded.

Result

At $\Gamma_{\mathrm{tot}} = 3$ pmol cm^{-2} probes are isolated (spacing 7.4 nm) and number 51 fmol per electrode. The electrode-scale signal is proportional to Γ_{tot} (Part D), so every prediction of absolute current inherits its uncertainty, but the cleavage readout of Eq. (I.1) does not (Part F).

III. Part B: Where the MB Label Sits and How Fast It Exchanges Electrons

Overview

Objective. To determine the distribution of the distance d of the MB label from the gold and the dependence of the electron-transfer rate on d. **Result.** Modelling the peptide as a freely jointed chain gives a half-Gaussian distribution of d, with the label most probably within about 1 nm of the PEG6 tip. The electron-transfer rate constant decreases by a factor of ten for every 2.30 Å (0.23 nm) of additional distance, so that only labels close to the gold exchange electrons at the rate of the square wave.

Assumptions

- **The peptide behaves as a freely jointed chain (FJC).** A standard model for a flexible, unstructured polymer, supported for short unfolded peptides [3]. A modelling choice, not a measurement on this peptide *(assumed)*.
- **The PEG6 spacer sets a floor.** The peptide is treated as rooted at the PEG6 tip, at height d_{PEG} above the gold, and cannot pass below it.
- **Isolated probe.** Justified by the spacing of Part A; not tested on the real surface *(assumed)*.
- **Electron transfer is tunnelling through the tether,** with a rate decaying exponentially with distance. The decay constant is taken from alkanethiols; no PEG-specific value is available *(assumed)*.

Equations and derivation

Chain statistics. The peptide has $N_{\mathrm{res}} = 10$ residues, each adding $l_0 \approx 0.38$ nm of contour length, and a Kuhn length $b \approx 0.76$ nm (two residues) [3]: the length over which the chain has effectively forgotten its direction. The chain is therefore equivalent to

$$N_K = \frac{N_{\mathrm{res}}\, l_0}{b} = \frac{10 \times 0.38}{0.76} = 5 \tag{III.1}$$

independent rigid segments of length b, each pointing in a random direction. Take z as the height above the PEG6 tip. A segment whose direction is uniformly random over the sphere has a height component $X = b\cos\theta$. The fraction of the sphere's area between polar angles θ and $\theta + \mathrm{d}\theta$ is $\frac{1}{2}\sin\theta\,\mathrm{d}\theta = \frac{1}{2}|\mathrm{d}(\cos\theta)|$, so $u = \cos\theta$ is uniformly distributed on $[-1, 1]$ with density $\frac{1}{2}$. Therefore

$$\langle X\rangle = b\int_{-1}^{1} u\,\tfrac{1}{2}\,\mathrm{d}u = 0, \qquad \langle X^2\rangle = b^2\int_{-1}^{1} u^2\,\tfrac{1}{2}\,\mathrm{d}u = b^2\cdot\tfrac{1}{2}\cdot\tfrac{2}{3} = \frac{b^2}{3}.$$

The end height is $S_N = X_1 + \cdots + X_N$, the sum of $N = N_K$ independent, identically distributed terms, with mean $\langle S_N\rangle = N\langle X\rangle = 0$ and, expanding the square, $\langle S_N^2\rangle = \sum_i \langle X_i^2\rangle + \sum_{i\neq j}\langle X_i\rangle\langle X_j\rangle = Nb^2/3 + 0$ (the cross terms factorise because the segments are independent, and vanish because $\langle X\rangle = 0$). The *central limit theorem* states that if the X_i have mean μ and variance s^2, then

$$\lim_{N\to\infty} \Pr\left[\frac{S_N - N\mu}{s\sqrt{N}} \le x\right] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-y^2/2}\,\mathrm{d}y, \tag{III.2}$$

that is, S_N is approximately normal with mean $N\mu$ and variance Ns^2. Here $\mu = 0$ and $s^2 = b^2/3$, so S_N is approximately normal with mean zero and variance

$$\sigma_z^2 = \frac{N_K\, b^2}{3} = \frac{\langle R^2\rangle}{3}, \qquad \langle R^2\rangle = N_K b^2 = 5\times 0.76^2 = 2.89\ \mathrm{nm}^2, \qquad \sigma_z = \sqrt{2.89/3} = 0.98\ \mathrm{nm}. \tag{III.3}$$

Here $\langle R^2\rangle$ is the mean square end-to-end distance. With the end-to-end vector $\mathbf{R} = \sum_i \mathbf{b}_i$, $\langle R^2\rangle = \sum_i |\mathbf{b}_i|^2 + \sum_{i\neq j}\langle \mathbf{b}_i\cdot\mathbf{b}_j\rangle = N_K b^2 + 0$: each segment has $|\mathbf{b}_i|^2 = b^2$, and the cross terms vanish because $\langle \mathbf{b}_i\cdot\mathbf{b}_j\rangle = \langle\mathbf{b}_i\rangle\cdot\langle\mathbf{b}_j\rangle = 0$ for independent isotropic segments. The factor 3 arises because, by isotropy, $\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle$ and these sum to $\langle R^2\rangle$. The same argument gives the sideways displacement: with x and y each of variance σ_z^2, the root-mean-square distance of the label from the vertical through its anchor is $\sqrt{\langle x^2\rangle + \langle y^2\rangle} = \sqrt{2}\,\sigma_z = 1.4$ nm, as used in Part A. Five segments is few, so the approximation is checked directly: the exact distribution of the sum of five such terms differs from the Gaussian by a maximum of 0.007 in cumulative probability (Figure 6B), so the Gaussian is an excellent description.

Reference height: the PEG6 spacer. The spacer is six ethylene-glycol units of length $l_{\mathrm{EO}} = 0.35$ nm, or $n_{K,\mathrm{PEG}} = 6\times 0.35/0.76 = 2.76$ Kuhn segments. For a chain in a good solvent, Flory's argument balances two contributions to the free energy in units of k_BT: the elastic cost of stretching the chain to an end-to-end distance R, $R^2/(n_Kb^2)$, and the repulsion between segments, which for n_K segments in a volume $\sim R^3$ is $\sim b^3n_K^2/R^3$. Minimising the sum with respect to R: $\frac{d}{dR}\left[R^2/(n_Kb^2)\right] = 2R/(n_Kb^2)$ and $\frac{d}{dR}\left[b^3n_K^2R^{-3}\right] = -3b^3n_K^2R^{-4}$, so the condition $\frac{d}{dR}(\cdot) = 0$ gives $2R/(n_Kb^2) = 3b^3n_K^2/R^4$. Multiplying both sides by $n_Kb^2R^4/2$ gives $R^5 = \frac{3}{2}\,b^5n_K^3$, hence $R = (3/2)^{1/5}\,n_K^{3/5}\,b \approx 1.08\,n_K^{3/5}b$; to scaling accuracy the numerical factor is dropped and

$$d_{\mathrm{PEG}} = R_F = n_{K,\mathrm{PEG}}^{3/5}\,b = 2.76^{0.6}\times 0.76 = 1.40\ \mathrm{nm}. \tag{III.4}$$

This is an extrapolation of a long-chain law to a chain of under three segments and is therefore uncertain *(assumed)*. The ideal-chain value $\sqrt{n_{K,\mathrm{PEG}}}\,b = 1.26$ nm and the fully stretched value $6\times 0.35 = 2.1$ nm bracket it, and this range is carried into the sensitivity analysis (Part H).

Distribution of d. A normal distribution of mean zero and variance σ_z^2 has probability density

$$f(z) = \frac{1}{\sigma_z\sqrt{2\pi}}\exp\!\left(-\frac{z^2}{2\sigma_z^2}\right), \qquad \int_{-\infty}^{\infty} f\,\mathrm{d}z = 1. \tag{III.5}$$

The label height relative to the tip is z, but the gold prevents the chain from lying below the tip, so the label is at $d = d_{\mathrm{PEG}} + |z|$. Let $u = d - d_{\mathrm{PEG}} \ge 0$. The event that $|z|$ lies between u and $u + \mathrm{d}u$ consists of z lying in $[u, u+\mathrm{d}u]$ or in $[-u-\mathrm{d}u, -u]$; these have equal probability, $f(u)\,\mathrm{d}u$, because f is even. Hence the density of u is $2f(u)$:

$$P(d) = 2f(d - d_{\mathrm{PEG}}) = \frac{2}{\sigma_z\sqrt{2\pi}}\,e^{-(d-d_{\mathrm{PEG}})^2/2\sigma_z^2} = \sqrt{\frac{2}{\pi}}\,\frac{1}{\sigma_z}\,\exp\!\left[-\frac{(d-d_{\mathrm{PEG}})^2}{2\sigma_z^2}\right], \qquad d \ge d_{\mathrm{PEG}}, \tag{III.6}$$

using $2/\sqrt{2\pi} = \sqrt{2/\pi}$. It is normalised because $\int_{d_{\mathrm{PEG}}}^{\infty} P\,\mathrm{d}d = 2\int_0^{\infty} f\,\mathrm{d}u = \int_{-\infty}^{\infty} f\,\mathrm{d}z = 1$ (f is even, so half of the total area lies on each side of zero; the last integral is evaluated in Appendix A, Eq. X.2). The mean distance follows from $\int_0^{\infty} u\,e^{-u^2/2\sigma_z^2}\mathrm{d}u = \sigma_z^2$ (Eq. X.4, Appendix A): $\langle d\rangle = d_{\mathrm{PEG}} + \frac{2}{\sigma_z\sqrt{2\pi}}\sigma_z^2 = d_{\mathrm{PEG}} + \sigma_z\sqrt{2/\pi} = 2.18$ nm. The chain cannot extend beyond its contour length, so the distribution is cut off at $d_{\max} = d_{\mathrm{PEG}} + N_{\mathrm{res}}l_0 = 5.2$ nm; the probability of exceeding it, $1 - \mathrm{erf}\left[(d_{\max} - d_{\mathrm{PEG}})/(\sqrt{2}\sigma_z)\right]$ (Eq. X.5, Appendix A), is 1.1×10^{-4}, which is negligible.

Checking the folding. Folding a Gaussian is not the same as a chain that is prevented from entering the region below its root. Two exact alternatives were simulated (Figure 6C; 4×10^5 random chains). Folding the exact end-height distribution gives a mean d of 2.19 nm, indistinguishable from the half-Gaussian (2.18 nm), and a predicted signal within 3% of it. Requiring that *no* segment of the chain passes below the tip is stricter: only 25% of random chains satisfy it, the distribution shifts to larger d (mean 2.43 nm), and the predicted peak current falls

by 38% (9.9 against 15.9 nA). The half-Gaussian is therefore the central estimate, and the stricter wall model is carried into the sensitivity analysis as a lower bound on the geometry.

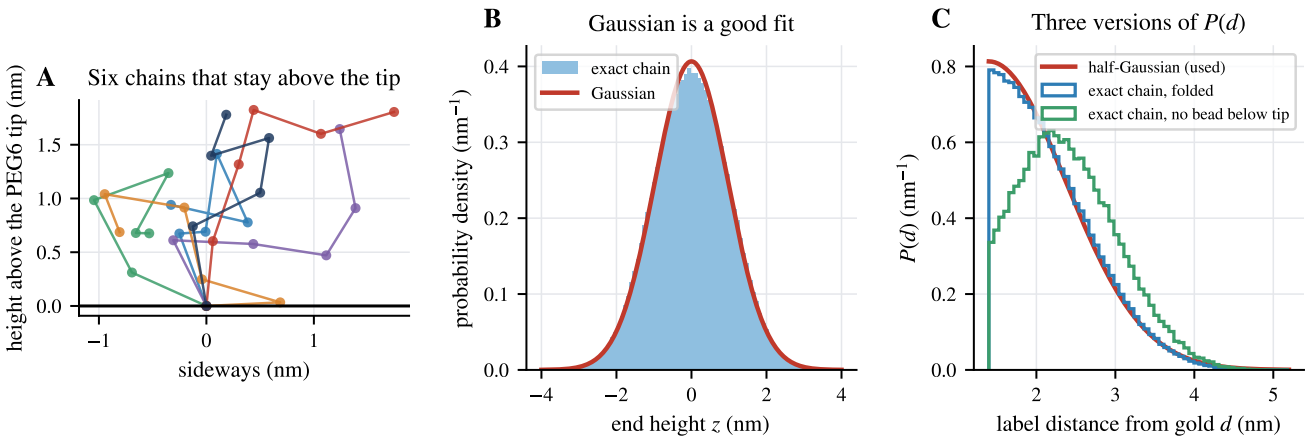


Figure 6: **A:** six random five-segment chains that stay above the PEG6 tip (horizontal line), showing the spread of label heights. **B:** the exact distribution of the end height (histogram) and the Gaussian of Eq. (III.3) (line); they agree to within 0.007 in cumulative probability. **C:** the three versions of $P(d)$ compared: the half-Gaussian used in the model, the folded exact chain, and the exact chain constrained so that no segment passes below the tip.

Electron-transfer rate against distance. MB exchanges electrons with the gold by quantum-mechanical tunnelling through the tether. For an electron of energy E_e meeting a barrier of height $\Phi > E_e$ and width d, the Schrödinger equation inside the barrier, $-\frac{\hbar^2}{2m_e}\psi'' + \Phi\psi = E_e\psi$, becomes $\psi'' = \gamma^2\psi$ with $\gamma = \sqrt{2m_e(\Phi - E_e)}/\hbar$, whose decaying solution is $\psi \propto e^{-\gamma x}$. The transmission probability is the ratio of $|\psi|^2$ at the far and near faces, $e^{-2\gamma d}$ (writing Φ for the barrier height measured from E_e). The electron-transfer rate is proportional to it, and the rate constant k^0 (the rate at which an MB molecule exchanges electrons when it is at its formal potential, in s^{-1}) inherits this:

$$k^0(d) = k^0(0)\, e^{-\beta d}, \qquad \beta = 2\gamma = \frac{2\sqrt{2m_e\Phi}}{\hbar}. \tag{III.7}$$

Numerically, for $\Phi = 1$ eV$= 1.602\times10^{-19}$ J, $\sqrt{2m_e\Phi} = \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}} = 5.40\times10^{-25}$ kg m s^{-1}, so $\beta = 2 \times 5.40 \times 10^{-25}/(1.055 \times 10^{-34}) = 1.02 \times 10^{10}$ m^{-1}, i.e. $\beta = 1.025\sqrt{\Phi/\mathrm{eV}}$ Å^{-1}. A barrier of about 1 eV gives $\beta = 1.0$ Å$^{-1} = 10$ nm^{-1}, the accepted value for tunnelling through alkane chains [4]; no PEG-specific value is available, so this is an assumed value *(assumed)*. The anchoring measurement is that of Dauphin-Ducharme *et al.* [1]: MB on a six-carbon linker, with the label 8 Å from the gold, has $k^0 = 5.0 \times 10^4$ s^{-1}. Eq. (III.7) extrapolated back to zero distance gives $k^0(0) = 5.0 \times 10^4\, e^8 = 1.5 \times 10^8$ s^{-1}, and evaluated at the PEG6 tip,

$$k_{\mathrm{ref}} = k^0(d_{\mathrm{PEG}}) = 1.5 \times 10^8 \times e^{-10\times1.40} = 1.5 \times 10^8 \times e^{-13.98} = 1.5 \times 10^8 \times 8.5 \times 10^{-7} = 127\ \mathrm{s}^{-1}. \tag{III.8}$$

Because k^0 falls tenfold per 0.23 nm, a label 0.5 nm above the tip is already ~ 150 times slower. Which labels contribute at a square-wave frequency of 10 to 20 Hz is therefore determined not by $P(d)$ alone but by the balance shown in Figure 7, quantified in Parts C and D.

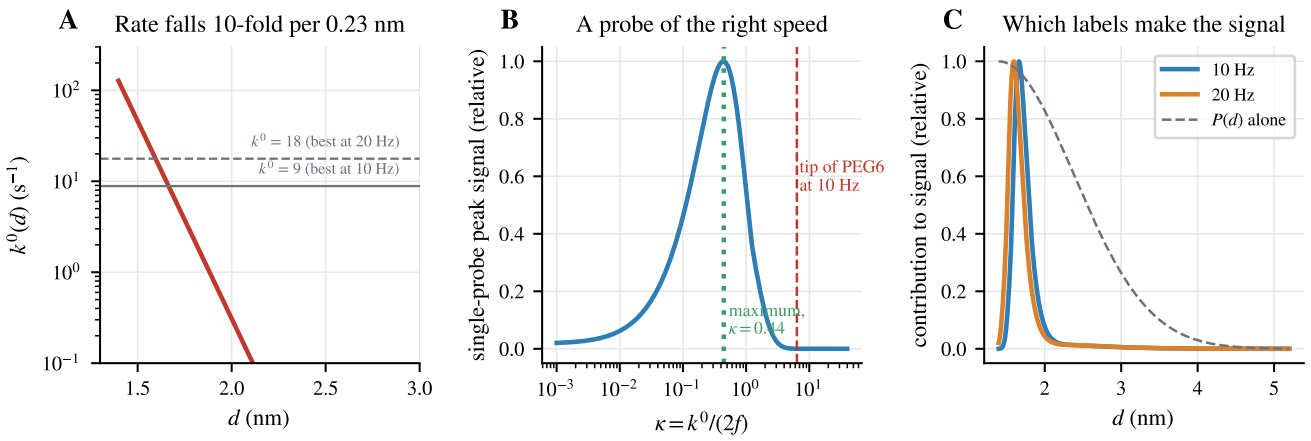


Figure 7: **A:** the rate constant against distance (Eq. III.7); horizontal lines mark the rate at which a single probe gives its largest signal at 10 and 20 Hz (Part C). **B:** the single-probe peak signal against $\kappa = k^0/(2f)$, the ratio of the label's rate to the pulse rate (defined in Part C); the maximum is at $\kappa = 0.44$, and a label at the PEG6 tip has $\kappa = 6.4$ at 10 Hz, too fast. **C:** the contribution of each distance to the total signal is the product of the probability of finding a label there (dashed) and its single-probe signal. It is concentrated in a narrow band just above the tip, not at the most probable distance.

Result

Labels are spread over $d = 1.40$ to about 5.2 nm with mean 2.18 nm (Eq. III.6), and their electron-transfer rate constants span more than fifteen orders of magnitude. At the square-wave frequencies used, only a narrow band, a few tenths of a nanometre above the PEG6 tip (Part D), produces signal.

IV. Part C: Square-Wave Voltammetry and the Signal from One Probe

Overview

Objective. To determine the current produced by a single MB label at a fixed distance from the gold under a square-wave potential program. **Result.** The reduced fraction of MB relaxes exponentially towards the value set by the applied voltage at a rate set by the electron-transfer rate constant. The current sampled at the end of each pulse is proportional to the displacement from equilibrium at that instant, and is therefore large only when the rate constant is comparable to the pulse rate. The single-probe signal has a maximum at $\kappa \approx 0.44$ (Figure 11B).

Assumptions

- **Butler–Volmer kinetics for surface-confined MB.** The standard textbook model for a redox species attached to an electrode [13]; not specific to this device and not in question.
- **Constant voltage within each pulse.** True by construction of the square wave; it is what makes the equation below exactly solvable.
- **Sampling at the end of each pulse.** How the potentiostat samples the current affects the predictions at fast rates (Part E).
- **Two-electron exchange in one step,** with $\alpha = 0.37$, $n_e = 2$ [1,6] *(established).* The intermediate one-electron step of MB is not resolved in the model (the same simplification as the source [1]).
- **Room temperature,** $T = 298.15$ **K.** The wound is at about 310 K; the measurement temperature has not been set *(assumed).*

Why square-wave voltammetry

Any electrode voltage change also charges the electrical double layer at the gold surface, which acts as a capacitor of about $C_{\mathrm{dl}} = 20\ \mu\mathrm{F\,cm^{-2}}$ (a typical value for gold in aqueous electrolyte, assumed here), so 0.34 μF on the 0.017 $\mathrm{cm^2}$ electrode. This charging current is not the signal, and in cyclic voltammetry (CV), where the voltage is swept steadily, it is present all the time. The size of the wanted signal relative to it can be estimated. For a surface-bound couple in equilibrium with the electrode the reduced fraction is given by the Nernst equation, $\theta_{\mathrm{red}} = 1/(1+u)$ with $u = e^{\xi}$ and ξ as in Eq. (IV.2). In cyclic voltammetry (CV) the voltage is swept at a rate v, so $E = E_0 + vt$ and the current is

$$i = n_e F A \Gamma_{\mathrm{tot}} \frac{d\theta_{\mathrm{red}}}{dt} = n_e F A \Gamma_{\mathrm{tot}}\, v\, \frac{d\theta_{\mathrm{red}}}{dE}.$$

Differentiating $\theta_{\mathrm{red}} = (1+u)^{-1}$ with $du/dE = u\,d\xi/dE = u\,n_eF/(RT)$:

$$\frac{d\theta_{\mathrm{red}}}{dE} = -\frac{1}{(1+u)^2}\frac{du}{dE} = -\frac{n_eF}{RT}\,\frac{u}{(1+u)^2},$$

so $|i| = \dfrac{n_e^2F^2vA\Gamma_{\mathrm{tot}}}{RT}\,\dfrac{u}{(1+u)^2}$. The function $u/(1+u)^2$ has derivative $(1-u)/(1+u)^3$, so it is maximal at $u=1$ (that is, at $E = E^{0\prime}$), where it equals $\frac{1}{4}$, and the peak current is $i_p = n_e^2F^2vA\Gamma_{\mathrm{tot}}/(4RT)$. The charging current of a capacitor $C_{\mathrm{dl}}A$ swept at rate v is $i_c = C_{\mathrm{dl}}Av$ (from $q = CE$). Their ratio is independent of v:

$$\frac{i_p}{i_c} = \frac{n_e^2F^2\Gamma_{\mathrm{tot}}}{4RT\,C_{\mathrm{dl}}} = \frac{4\times(96485)^2\times 3\times 10^{-12}}{4\times 8.3145\times 298.15\times 20\times 10^{-6}} = 0.56, \tag{IV.1}$$

so the peak sits on a charging plateau of comparable size (Figure 8B), and any drift in the plateau is a drift in the apparent peak.

In SWV the voltage is held constant during each pulse and the current is recorded only at its end. After a voltage jump of size ΔE the charge q on the capacitor obeys $\Delta E = R_u\,dq/dt + q/C$. Let $y = q - C\Delta E$; then $dy/dt = dq/dt = (\Delta E - q/C)/R_u = -y/(R_uC)$, whose solution is $y = y_0e^{-t/(R_uC)}$ with $y_0 = q(0) - C\Delta E = -C\Delta E$. Hence $q = C\Delta E(1 - e^{-t/(R_uC)})$ and the charging current is $i = dq/dt = (\Delta E/R_u)\,e^{-t/(R_uC)}$, where R_u is the resistance of the solution between the electrode and reference (taken as 1000 Ω, an assumed value). With $R_uC = 0.34$ ms, by the end of a 10 Hz half-pulse (50 ms) the charging current has fallen by a factor of e^{-147}. Meanwhile the wanted Faradaic current, which decays on the much slower time scale of the electron-transfer kinetics, is still large. In addition, taking the *difference* between the forward and reverse currents cancels remaining slowly varying background. Both steps remove background that CV retains, giving SWV its greater sensitivity for a surface-bound probe [5,11].

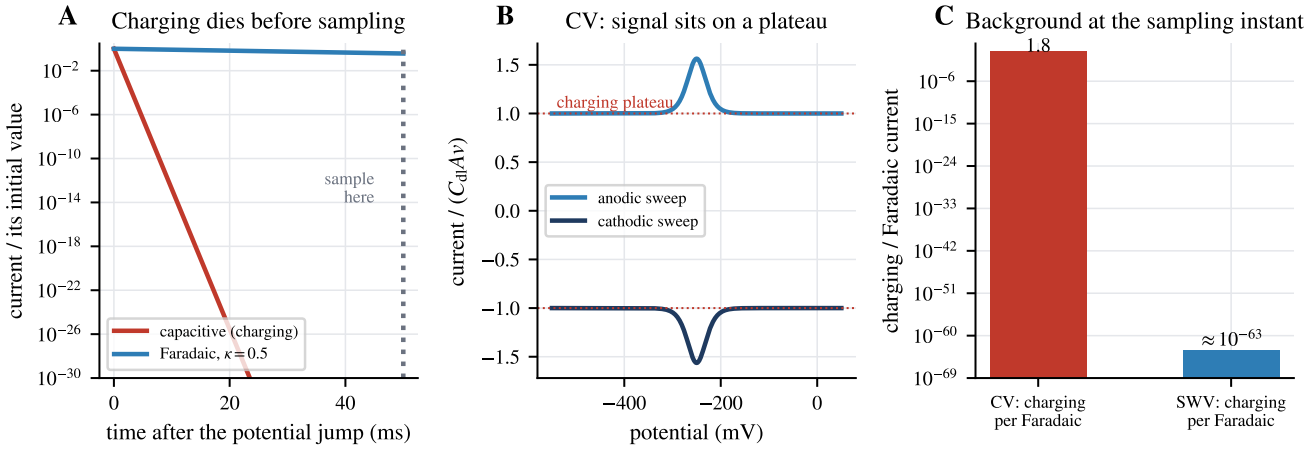


Figure 8: **A:** after a voltage jump the charging current (red) dies out in fractions of a millisecond, whereas the Faradaic current of a label with $\kappa = 0.5$ (blue) persists to the end of the pulse, when the current is sampled. The values of C_{dl} and R_u are assumed. **B:** in cyclic voltammetry the surface-bound peak sits on a charging plateau of the same order (Eq. IV.1). **C:** charging current per unit of Faradaic current at the moment of measurement.

Equations and derivation

Rate constants. Let Γ_{red} and Γ_{ox} be the amounts of MB per unit area in the reduced and oxidised forms ($\mathrm{mol\,cm^{-2}}$), with $\Gamma_{\mathrm{tot}} = \Gamma_{\mathrm{red}} + \Gamma_{\mathrm{ox}}$ fixed. At applied voltage E, the Butler–Volmer model gives first-order rate constants for reduction (k_f) and oxidation (k_b):

$$k_f = k^0\exp\!\Big[-\alpha\,\xi\Big], \qquad k_b = k^0\exp\!\Big[(1-\alpha)\,\xi\Big], \qquad \xi = \frac{n_eF\,(E-E^{0\prime})}{RT}, \tag{IV.2}$$

where $F = 96485\ \mathrm{C\,mol^{-1}}$ is the Faraday constant, $R = 8.3145\ \mathrm{J\,mol^{-1}K^{-1}}$ the gas constant, α the transfer coefficient (the fraction of the applied voltage that helps the reaction go in one direction) and ξ the overpotential in thermal units. Eq. (IV.2) follows from transition-state theory, in which a rate is proportional to $\exp(-\Delta G^{\ddagger}/RT)$ with $\Delta G^{\ddagger}$ the free-energy barrier. Changing the electrode voltage by $E - E^{0\prime}$ changes the free energy of the transferred electrons by $n_eF(E-E^{0\prime})$. A fraction α of this change raises the barrier for reduction and the remaining

fraction $1-\alpha$ lowers the barrier for oxidation: $\Delta G_f^\ddagger = \Delta G_0^\ddagger + \alpha n_e F(E-E^{0\prime})$ and $\Delta G_b^\ddagger = \Delta G_0^\ddagger - (1-\alpha) n_e F(E-E^{0\prime})$. At $E = E^{0\prime}$ both barriers equal $\Delta G_0^\ddagger$, so both rates equal $k^0 \propto e^{-\Delta G_0^\ddagger/RT}$, and dividing out gives Eq. (IV.2). Positive of $E^{0\prime}$, k_b grows and k_f shrinks, favouring the oxidised form, and vice versa. At equilibrium the two rates balance, $k_f\Gamma_{\mathrm{ox}} = k_b\Gamma_{\mathrm{red}}$, so $\Gamma_{\mathrm{red}}/\Gamma_{\mathrm{ox}} = k_f/k_b = \exp[-\alpha\xi - (1-\alpha)\xi] = e^{-\xi}$, the Nernst equation, independent of α and k^0.

Governing equation. Reduced MB is formed from oxidised at rate $k_f\Gamma_{\mathrm{ox}}$ and lost at rate $k_b\Gamma_{\mathrm{red}}$:

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f\Gamma_{\mathrm{ox}} - k_b\Gamma_{\mathrm{red}} = k_f\Gamma_{\mathrm{tot}} - K\,\Gamma_{\mathrm{red}}, \qquad K \equiv k_f + k_b. \tag{IV.3}$$

Solution. Within a pulse E, hence k_f and k_b, is constant, so Eq. (IV.3) is a first-order linear equation with constant coefficients, $\dot\Gamma_{\mathrm{red}} + K\Gamma_{\mathrm{red}} = k_f\Gamma_{\mathrm{tot}}$. Multiply both sides by e^{Kt}. By the product rule, $\frac{d}{dt}\big(\Gamma_{\mathrm{red}}e^{Kt}\big) = \dot\Gamma_{\mathrm{red}}e^{Kt} + K\Gamma_{\mathrm{red}}e^{Kt}$, which is exactly the left-hand side, so

$$\frac{d}{dt}\big(\Gamma_{\mathrm{red}}\,e^{Kt}\big) = k_f\Gamma_{\mathrm{tot}}\,e^{Kt}.$$

Integrate from $t = 0$ to $t = \tau$. The left side integrates to $\Gamma_{\mathrm{red}}(\tau)e^{K\tau} - \Gamma_{\mathrm{red}}(0)$ and the right side to $k_f\Gamma_{\mathrm{tot}}\,(e^{K\tau}-1)/K$:

$$\Gamma_{\mathrm{red}}(\tau)\,e^{K\tau} - \Gamma_{\mathrm{red}}(0) = \frac{k_f\Gamma_{\mathrm{tot}}}{K}\big(e^{K\tau}-1\big).$$

Add $\Gamma_{\mathrm{red}}(0)$ and divide by $e^{K\tau}$:

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{red}}(0)\,e^{-K\tau} + \frac{k_f\Gamma_{\mathrm{tot}}}{K}\big(1-e^{-K\tau}\big).$$

The quantity $\Gamma_{\mathrm{eq}} \equiv k_f\Gamma_{\mathrm{tot}}/K$ is the value at which $d\Gamma_{\mathrm{red}}/dt = 0$ in Eq. (IV.3), the equilibrium coverage at this voltage. Substituting it and collecting the terms in $e^{-K\tau}$, $\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{red}}(0)e^{-K\tau} + \Gamma_{\mathrm{eq}} - \Gamma_{\mathrm{eq}}e^{-K\tau}$, that is

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + \big[\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}\big]\,e^{-K\tau}. \tag{IV.4}$$

The coverage relaxes exponentially to equilibrium with time constant $1/K$. This solution is exact, and agrees with a numerical ODE solver to 5×10^{-13} (Figure 10A,B).

Current. The current is the charge transferred per second: each mole converted transfers n_eF coulombs, the net rate of conversion per unit area is $k_b\Gamma_{\mathrm{red}} - k_f\Gamma_{\mathrm{ox}}$ ($\mathrm{mol\,cm^{-2}s^{-1}}$), and the area is A, so (in $\mathrm{C\,mol^{-1}}\times\mathrm{mol\,cm^{-2}s^{-1}}\times\mathrm{cm^2}$ = A)

$$I = n_eFA\,\big(k_b\Gamma_{\mathrm{red}} - k_f\Gamma_{\mathrm{ox}}\big). \tag{IV.5}$$

Substitute $\Gamma_{\mathrm{ox}} = \Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}$ and collect the terms in Γ_{red}:

$$k_b\Gamma_{\mathrm{red}} - k_f(\Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}) = (k_b + k_f)\Gamma_{\mathrm{red}} - k_f\Gamma_{\mathrm{tot}} = K\Big(\Gamma_{\mathrm{red}} - \frac{k_f\Gamma_{\mathrm{tot}}}{K}\Big) = K\big(\Gamma_{\mathrm{red}} - \Gamma_{\mathrm{eq}}\big).$$

Now insert the solution, Eq. (IV.4), which gives $\Gamma_{\mathrm{red}}(\tau) - \Gamma_{\mathrm{eq}} = [\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}]e^{-K\tau}$:

$$I = n_eFA\,K\,\big[\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}\big]\,e^{-K\tau}. \tag{IV.6}$$

The current is K times the displacement from equilibrium, and the displacement decays as $e^{-K\tau}$. Eqs. (IV.5) and (IV.6) are algebraically identical, but they are not numerically equal in floating-point arithmetic. When $K\tau$ is large, each term of Eq. (IV.5) is of order $k\Gamma_{\mathrm{tot}}$ and their difference is smaller than the rounding error of either. Evaluating Eq. (IV.5) therefore returns numerical noise instead of the physical result for $\kappa \gtrsim 15$, which is a purely numerical artefact. Using Eq. (IV.6) removes the cancellation (Figure 10C): the two forms agree to better than 10^{-8} relative where both are reliable, and the stable form continues correctly to arbitrarily fast rates.

The SWV sequence. At each staircase voltage E_{step} the electrode is held at $E_{\mathrm{step}} + E_{\mathrm{sw}}$ for $\tau = 1/(2f)$ (forward pulse), and then at $E_{\mathrm{step}} - E_{\mathrm{sw}}$ for another τ (reverse pulse), where f is the square-wave frequency (Figure 9A). The coverage at the end of one pulse is the starting coverage of the next (Eq. IV.4 applied twice in succession, Figure 9B). The net current is

$$\Delta I(E_{\mathrm{step}}) = I_{\mathrm{fwd}} - I_{\mathrm{rev}}, \tag{IV.7}$$

evaluated after five cycles at the starting potential so that the result is independent of the arbitrary initial coverage.

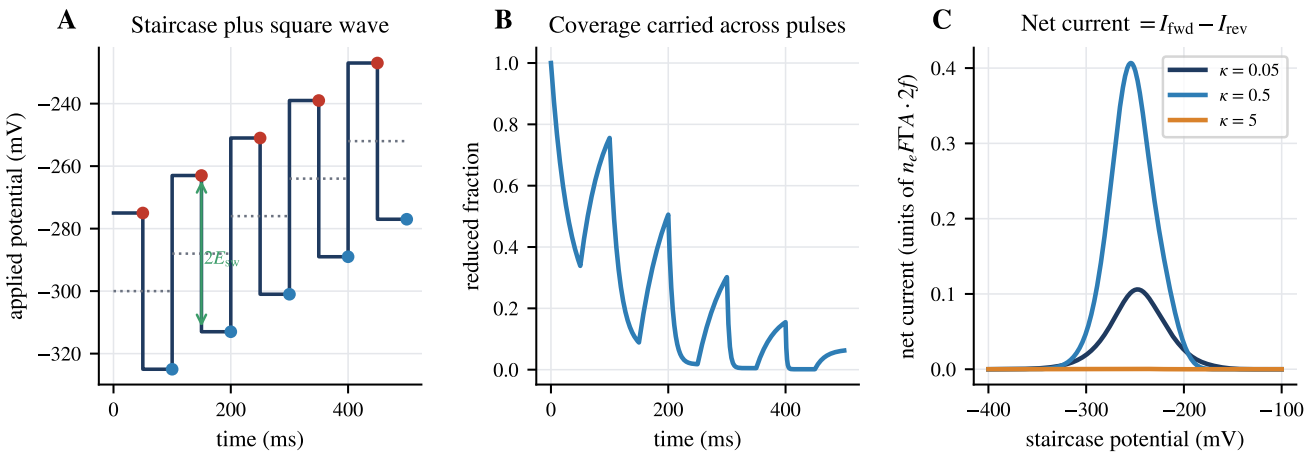


Figure 9: **A:** the applied voltage: a staircase (dotted) with a square wave of amplitude E_{sw} on top; the current is sampled at the end of each pulse (red dots: forward, blue dots: reverse). The staircase step is drawn larger than the real one for visibility. **B:** the reduced fraction of one probe over a few steps; it moves towards a different equilibrium in each pulse and carries over. **C:** the net current against staircase voltage for probes of three different speeds.

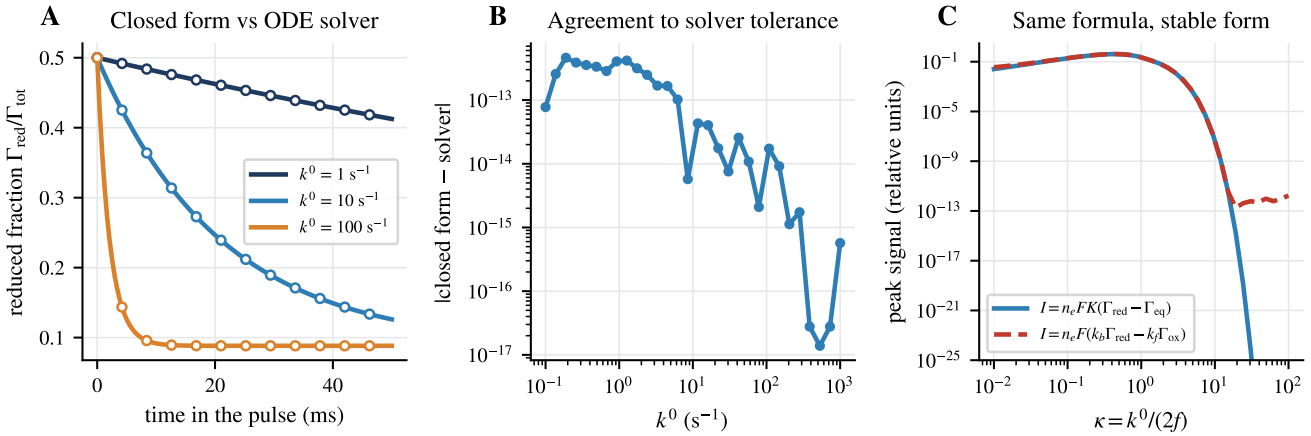


Figure 10: **A:** the reduced fraction over one pulse for three rate constants: Eq. (IV.4) (lines) against a numerical ODE solver (circles). **B:** their difference, at the level of the solver tolerance. **C:** peak signal against κ computed two ways. The stable form (Eq. IV.6, blue) continues smoothly; the two-term form (Eq. IV.5, red) fails beyond $\kappa \approx 15$ owing to cancellation.

The dimensionless rate and the best speed. Define

$$\kappa \equiv \frac{k^0}{2f} = k^0\,\tau, \tag{IV.8}$$

the ratio of the label's rate to the pulse rate. At $E = E^{0\prime}$, $\xi = 0$ and Eq. (IV.2) gives $k_f = k_b = k^0$, so $K = 2k^0$ and $K\tau = 2k^0\tau = 2\kappa$. Eq. (IV.6) says the current is K times a displacement that shrinks as $e^{-K\tau}$; the product $K\,e^{-K\tau}$ has derivative $e^{-K\tau}(1 - K\tau)$ with respect to K, so it is maximal where $K\tau = 1$, that is at $\kappa = \frac{1}{2}$. The full simulation, which includes the carry-over between pulses, places the maximum at $\kappa = 0.44$ (Figure 10C, Figure 7B), in agreement with the estimate. A label that exchanges much faster ($\kappa \gg 1$) reaches equilibrium within each pulse and its sampled current is nearly zero: physically, the electrons have already moved before the current is measured. A label much slower ($\kappa \ll 1$) responds negligibly. Numerically, the largest single-probe signal is $g_{max} = 0.41$ in units of $n_e F \cdot 2f$ per unit $A\Gamma_{tot}$.

Peak shape. For probes at $\kappa \lesssim 1$ the net peak is a single lobe at $E^{0\prime}$, with a width between 51 and 64 mV depending on κ; at larger κ the forward and reverse currents peak at different voltages and the net curve splits into two lobes

(Figure 11A). For comparison, a reversible, continuously-swept couple gives a peak of width $90.6/n_e = 45.3$ mV, derived in Part E.

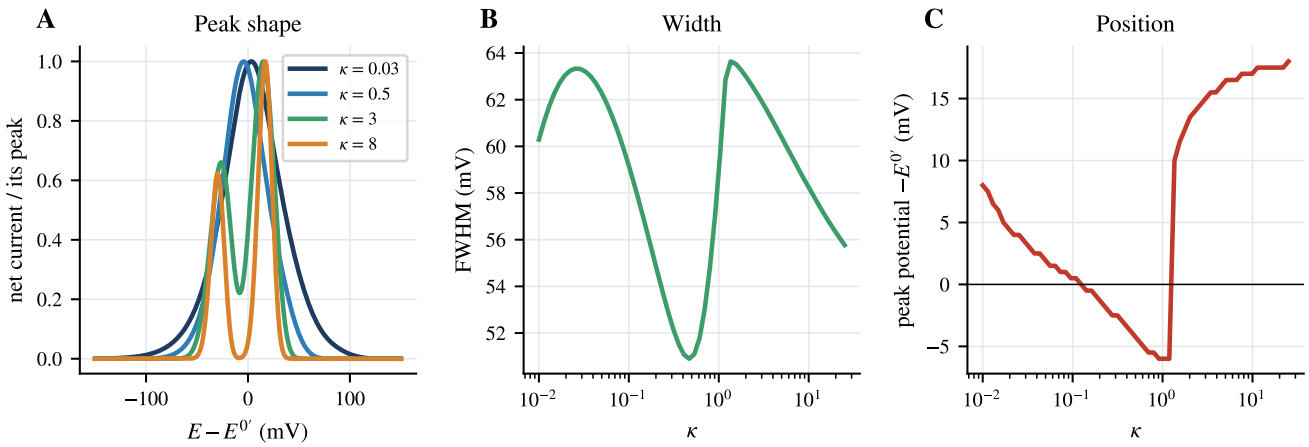


Figure 11: **A:** the shape of the net peak at four values of κ; at $\kappa = 8$ it has split into two lobes. **B:** full width at half maximum of the tallest lobe against κ. **C:** position of the peak relative to $E^{0\prime}$. The jump near $\kappa \approx 1.2$ is the change from one lobe to two.

Result

A single MB label gives a net peak at $E^{0\prime} = -0.25$ V whose height, in units of $n_e F \cdot 2f\, A\Gamma_{\text{tot}}$, is at most 0.41, reached at $\kappa = 0.44$, and falls to 2.7% of that at $\kappa = 3$ and to 0.09% at $\kappa = 5$. A label at the PEG6 tip of this device has $\kappa = 6.4$ at 10 Hz and 3.2 at 20 Hz, so it is faster than the optimum and contributes only 0.0083% of the maximum signal per probe at 10 Hz; this is what determines which labels give the signal (Part D).

V. Part D: The Signal from the Whole Electrode

Overview

Objective. To determine the total signal of the probe population and its dependence on the cleavage fraction. **Result.** Summation over the probes, weighted by $P(d)$, gives a single peak of about 15.9 nA at 10 Hz. It arises from a narrow band of labels ∼0.27 nm above the PEG6 tip. The height is exactly proportional to the number of intact probes; the position and width are independent of cleavage.

Assumptions

- **Probes are independent.** Justified by the spacing of Part A.
- **Cleaved probes give no signal, and intact probes are unaffected by their neighbours being cleaved.** Once the peptide is cut, the MB fragment leaves the surface (Figure 1B).
- **Cleavage is equally likely for a probe at any label height** d**.** An assumption *(assumed)*. It underlies the proportionality in Eq. (V.2) and is tested in Part F.

Equations and derivation

Distance average. A label at distance d has $k^0(d)$ (Eq. III.7), hence a net current per unit area and unit coverage $\Delta I_{\text{norm}}(E; k^0(d))$ from Eq. (IV.7). The electrode carries $(1 - F_k)\Gamma_{\text{tot}} A$ of intact probe, distributed over d according to $P(d)$:

$$\Delta I(E; F_k) = (1 - F_k)\,\Gamma_{\text{tot}} A \int_{d_{\text{PEG}}}^{d_{\max}} P(d)\,\Delta I_{\text{norm}}\big(E; k^0(d)\big)\,\mathrm{d}d. \tag{V.1}$$

The integral has no closed form and is evaluated by quadrature: with $n = 80$ equally spaced distances $d_j = d_{\text{PEG}} + j\,\Delta d$, $j = 0, \ldots, n-1$, $\Delta d = (d_{\max} - d_{\text{PEG}})/(n-1)$, the integral is replaced by the sum

$$\sum_{j=0}^{n-1} P(d_j)\,\Delta I_{\text{norm}}\big(E; k^0(d_j)\big)\,\Delta d;$$

at 10 and 20 Hz the result is converged to 0.00002% and 0.033% with the 80 points used (Figure 15A).

Proportionality to intact probe. Eq. (V.1) depends on F_k only through the prefactor, so

$$\Delta I(E; F_k) = (1 - F_k)\,\Delta I(E; 0) \quad \text{exactly}, \tag{V.2}$$

under the assumptions above. The peak height falls in direct proportion; the position and width do not change (Figure 12). The proportionality is exact by construction, and the code confirms it to machine precision.

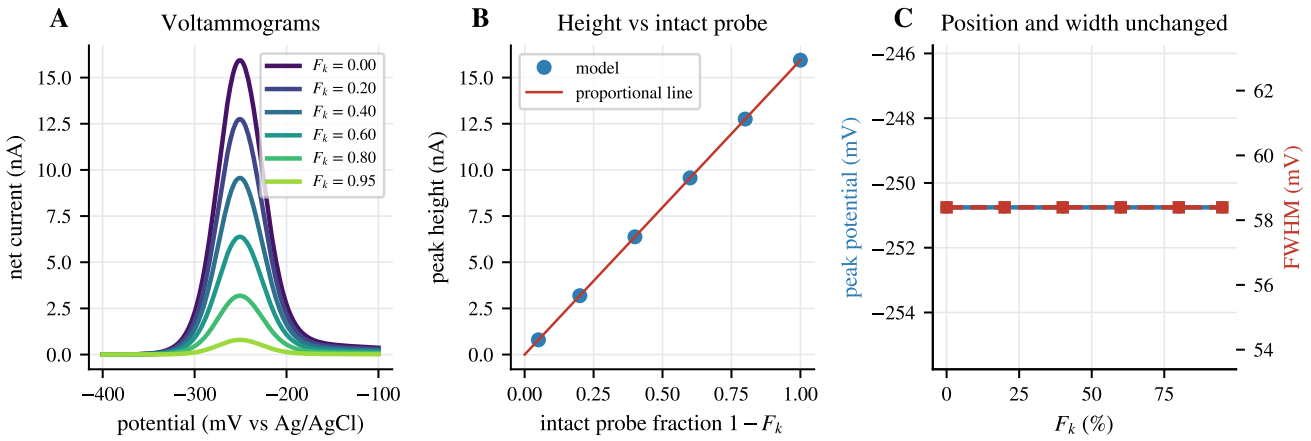


Figure 12: **A:** predicted voltammograms at six cleavage fractions. **B:** peak height against the intact fraction $1 - F_k$; the points lie on the proportional line. **C:** peak position and width do not change with F_k. The proportionality is exact by Eq. (V.2); what is not guaranteed is the assumption behind it (Part F).

Which labels give the signal. Write the single-probe response as $\Delta I_{\mathrm{norm}} = n_e F \cdot 2f\, g(\kappa)$, defining the dimensionless function g as in Part C, and let $\kappa_{\mathrm{tip}} = k_{\mathrm{ref}}/(2f)$ be the value of κ at the PEG6 tip. From Eq. (III.7), $\kappa(d) = \kappa_{\mathrm{tip}}\, e^{-\beta(d - d_{\mathrm{PEG}})}$, so $\ln\kappa = \ln\kappa_{\mathrm{tip}} - \beta(d - d_{\mathrm{PEG}})$ and

$$d = d_{\mathrm{PEG}} + \frac{1}{\beta}\big(\ln\kappa_{\mathrm{tip}} - \ln\kappa\big), \qquad \mathrm{d}d = -\frac{1}{\beta}\,\mathrm{d}\ln\kappa.$$

Substituting in Eq. (V.1) at the peak voltage, with the integration limits exchanged to absorb the sign,

$$\Delta I_{\mathrm{pk}} = n_e F \cdot 2f\, A\Gamma_{\mathrm{tot}}\, \frac{1}{\beta} \int P\big(d(\kappa)\big)\, g(\kappa)\, \mathrm{d}\ln\kappa.$$

The function $g(\kappa)$ is a narrow bump on the logarithmic axis (Figure 7B), whereas $P(d)$ varies slowly across it, so P can be evaluated at the centre of the bump, $d_\star$, where $\kappa(d_\star) = \kappa_{\max}$, i.e. $d_\star = d_{\mathrm{PEG}} + \beta^{-1}\ln(\kappa_{\mathrm{tip}}/\kappa_{\max})$, and taken outside the integral:

$$\Delta I_{\mathrm{pk}} \approx n_e F \cdot 2f\; A\Gamma_{\mathrm{tot}}\, \frac{P(d_\star)}{\beta}\, G, \qquad G = \int g(\kappa)\,\mathrm{d}\ln\kappa = 1.07, \qquad d_\star = d_{\mathrm{PEG}} + \frac{1}{\beta}\ln\frac{k_{\mathrm{ref}}}{2f\kappa_{\max}}. \tag{V.3}$$

With $k_{\mathrm{ref}} = 127\ \mathrm{s}^{-1}$, $\kappa_{\mathrm{tip}} = k_{\mathrm{ref}}/(2f) = 6.4$ at 10 Hz and 3.2 at 20 Hz, and $d_\star - d_{\mathrm{PEG}} = \beta^{-1}\ln(\kappa_{\mathrm{tip}}/\kappa_{\max}) = \frac{1}{10}\ln(6.4/0.44) = 0.27$ nm at 10 Hz. Thus at 10 Hz $d_\star = 1.66$ nm and at 20 Hz 1.60 nm, only 0.27 and 0.20 nm above the tip (Figure 13A,B). Eq. (V.3) evaluates as follows at 10 Hz. The density at the band centre is $P(d_\star) = \sqrt{2/\pi}\,\sigma_z^{-1} e^{-(d_\star - d_{\mathrm{PEG}})^2/(2\sigma_z^2)} = 0.7979/0.981 \times e^{-0.0368} = 0.784\ \mathrm{nm}^{-1}$, so $P(d_\star)/\beta = 0.784/10 = 0.0784$ (dimensionless). The prefactor is $n_e F \cdot 2f\, A\Gamma_{\mathrm{tot}} = 2 \times 96485 \times 20 \times 0.017 \times 3 \times 10^{-12} = 1.97 \times 10^{-7}$ A, and $\Delta I_{\mathrm{pk}} \approx 1.97 \times 10^{-7} \times 0.0784 \times 1.069 = 1.65 \times 10^{-8}$ A= 16.5 nA. The full quadrature gives 15.9 nA. Eq. (V.3) therefore reproduces the full calculation to within 3% (Figure 13C) and shows three things:

1. the peak is proportional to f: doubling the frequency doubles it (predicted ratio 2.04 for 20 against 10 Hz, a directly testable prediction);
2. it depends on the geometry only through $P(d_\star)/\beta$, not on the range of rate constants;
3. it is nearly independent of the anchoring value k_{ref}: changing k_{ref} shifts $d_\star$ but not the size of the bump. Scaling k_{ref} from 0.1 to 100 times its base value (127 s^{-1}) changes the peak to between 0.77 and 1.03 of its value (Figure 14B). This holds only while $d_\star$ stays within the range where probes exist: if the effective tip were much higher than assumed, so that k_{ref} fell by three decades, the signal would collapse (Part H).

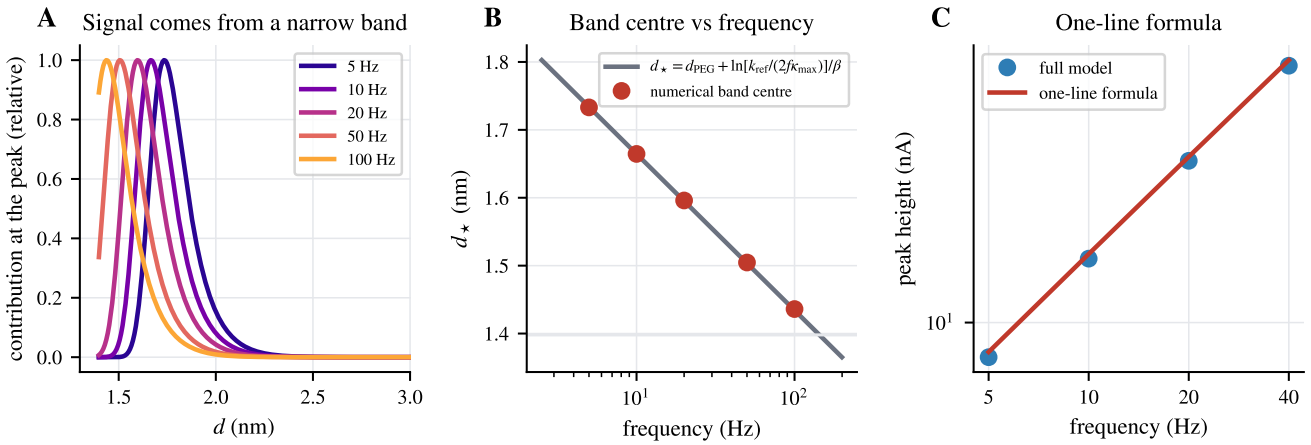


Figure 13: **A:** the contribution of each distance to the peak, at five frequencies; it is a narrow band. **B:** the centre of the band $d_\star$ against frequency: numerical (points) and Eq. (V.3) (line). **C:** peak height from the full model (points) and the one-line formula (line).

Frequency response. Because $\Delta I_{\mathrm{pk}} \propto f$ while the band lies above the tip, the predicted peak rises linearly with frequency from 1 Hz up to about $k_{\mathrm{ref}}/(2\kappa_{\max}) \approx 140$ Hz (the frequency at which $d_\star = d_{\mathrm{PEG}}$, from $\ln(k_{\mathrm{ref}}/2f\kappa_{\max}) = 0$ in Eq. (V.3)), where the band reaches the tip and the wall truncates it; the model's maximum is at about 398 Hz (Figure 14A). The operating range of 10 to 20 Hz is well below this, so the model predicts a larger signal at higher frequency (a fourfold gain from 10 to 40 Hz, 4.09); whether it is worth having depends on background and noise not treated here. Measuring peak height against frequency on a real electrode is the most direct available test of the whole rate model (Part E).

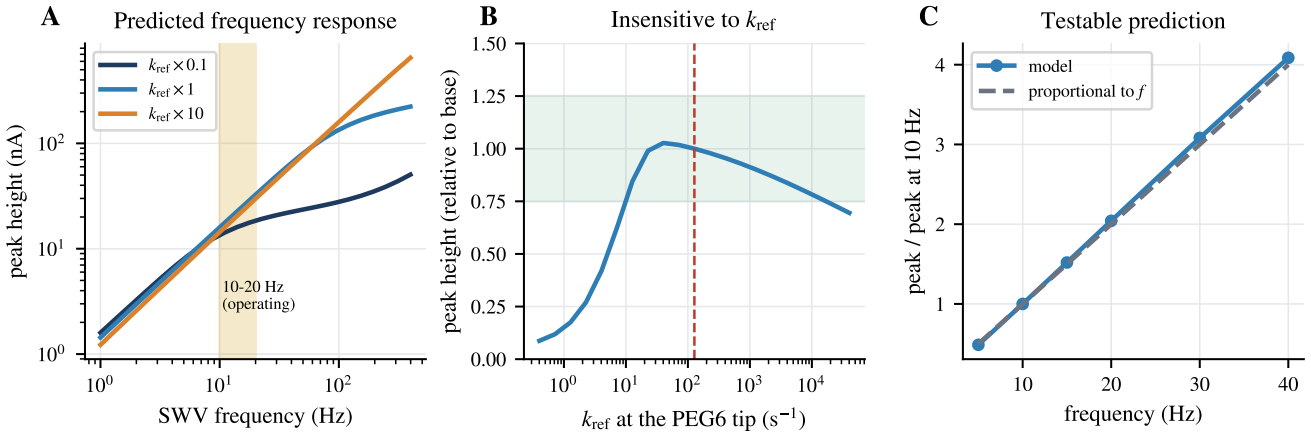


Figure 14: **A:** predicted peak height against frequency for three values of the anchoring rate k_{ref}; the shaded band is the 10 to 20 Hz operating range. **B:** peak height at 10 Hz against k_{ref}, relative to the base value: nearly flat over a range of about 1000. **C:** peak at each frequency relative to 10 Hz; the model gives a straight line through the origin.

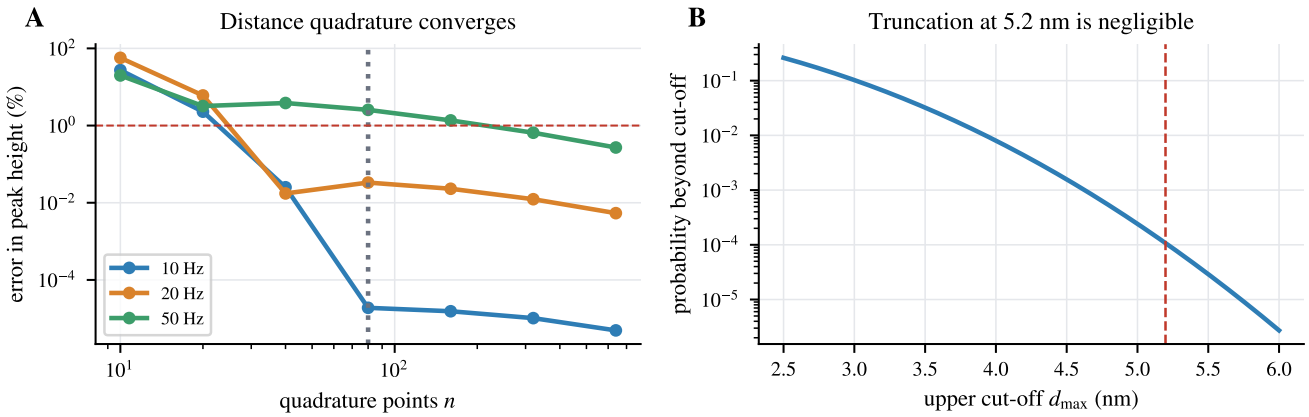


Figure 15: **A:** the error in the peak height against the number of quadrature points, against a 2000-point reference. The 80 points used are well converged at 10 and 20 Hz (at 50 Hz, 2%). **B:** the probability of a label beyond the upper cut-off $d_{\max}$ (dashed) is negligible.

Result

The model predicts a net peak of 15.9 nA at −251 mV, width 58 mV, at 10 Hz for $\Gamma_{\text{tot}} = 3$ pmol cm^{-2} and $A = 0.017$ cm^2 (about twice this at 20 Hz). Its height is exactly proportional to $(1 - F_k)$ and to Γ_{tot}, and is nearly independent of k_{ref}. The absolute value depends on the assumed inputs (Part H); the proportionality to $(1 - F_k)$ does not.

VI. Part E: Validation

Overview

Objective. To establish the correctness of the implementation and the agreement of the model with published behaviour. **Result.** The mathematics is verified directly to high precision, and the model reproduces the standard behaviour of square-wave voltammetry of a surface-bound couple. It does not reproduce the published high-rate voltammograms of Dauphin-Ducharme *et al.* [1], and the cause has not been established; this limits confidence in the absolute current but not in the cleavage readout. The model has not yet been compared with a measured voltammogram of this device.

Mathematical checks

The following checks are automated (`test_swv_model.py`, ten tests, all passing).

Check	Result
Closed-form solution, Eq. (IV.4), against a numerical ODE solver	agree to 5×10^{-13} over $k^0 = 0.1$ to 10^3 s^{-1}
Stable current, Eq. (IV.6), against the two-term form, Eq. (IV.5)	identical to better than 10^{-8} at $\kappa \lesssim 6$; the stable form stays finite and correct beyond $\kappa = 15$
Vectorised simulator against the reference simulator	identical for $k^0 = 0.5$, 10, 127 s^{-1}
$P(d)$ normalisation	integrates to 1 within 10^{-3}
Distance quadrature	converged to 0.00002% (10 Hz) and 0.033% (20 Hz) with 80 points
Proportionality to intact probe, Eq. (V.2)	exact to machine precision
Position of the single-probe maximum	$\kappa = 0.44$, in the expected range (0.2 to 1.0)
Default parameters	the default area and coverage of the distance-averaged function equal the module constants `A_ELEC` and `GAMMA_O`

The Laviron peak width and why it is not the target

The width of a *reversible* surface wave, swept continuously, is a useful reference, and it is the source of the 45.3 mV figure often quoted for a two-electron surface couple [2]. Its shape was obtained in Part C: the current is proportional

to $|d\theta_{\mathrm{red}}/dE| \propto \Psi(u) = u/(1+u)^2$ with $u = e^{\xi}$ and ξ as in Eq. (IV.2). Differentiating,

$$\frac{d\Psi}{du} = \frac{(1+u)^2 - 2u(1+u)}{(1+u)^4} = \frac{1-u}{(1+u)^3} = 0 \;\Rightarrow\; u = 1, \qquad \Psi_{\max} = \frac{1}{(1+1)^2} \cdot 1 = \frac{1}{4}.$$

Half the maximum is where $u/(1+u)^2 = \frac{1}{8}$, that is $8u = (1+u)^2$, i.e. $u^2 - 6u + 1 = 0$. By the quadratic formula $u_{\pm} = \left(6 \pm \sqrt{36-4}\right)/2 = 3 \pm 2\sqrt{2}$. The two roots are reciprocal ($u_+ u_- = 9 - 8 = 1$), so $\xi_{\pm} = \ln u_{\pm} = \pm \ln(3 + 2\sqrt{2}) = \pm 1.763$ and the width in ξ is $2\ln(3+2\sqrt{2}) = 3.525$. Since $\xi = n_e F(E - E^{0\prime})/RT$, a width $\Delta\xi$ corresponds to a width $\Delta\xi\, RT/(n_e F)$ in voltage:

$$\mathrm{FWHM} = \frac{3.525\, RT}{n_e F} = \frac{3.525 \times 25.69 \text{ mV}}{n_e} = \frac{90.6}{n_e} \text{ mV at } 25^{\circ}\mathrm{C}, \tag{VI.1}$$

(using $RT/F = 8.3145 \times 298.15/96485 = 25.69$ mV), which is 45.3 mV for $n_e = 2$. This is the width of a different experiment (a continuous sweep of a couple that stays at equilibrium) from the sampled-current square wave used here, where the peak is wider (51 to 64 mV, Part C) because the current is a non-equilibrium quantity sampled at a fixed time. A quantitative match to Eq. (VI.1) is therefore not expected; the difference in shape between the two (about 24% RMS at $\kappa = 10$) is a property of the two experiments.

Consistency with square-wave theory

The position of the single-probe maximum, $\kappa = 0.44$, means $k^0/f = 2\kappa = 2 \times 0.44 = 0.89$. The standard peak-current analysis of square-wave voltammetry for a surface-bound couple places the quasi-reversible maximum where the rate constant and the frequency are of the same order [1,5], which agrees. It also gives the same picture qualitatively: the signal vanishes for very slow and for very fast electron transfer, and the "optimum frequency" of a given label is $f \approx k^0$.

Comparison with Dauphin-Ducharme *et al.*

The most relevant published data are for methylene blue on a six-carbon linker on gold [1], whose Figure 4 shows net square-wave peaks at 20 to 1000 Hz. The paper reports $k^0 = 5.0 \times 10^4$ s^{-1} for the label at 8 Å, from a numerical simulation fitted to the data. At these rates $\kappa = k^0/(2f)$ is 250 at 100 Hz and 1250 at 20 Hz. The present model predicts essentially no net current there (Figure 16A: many orders of magnitude below the maximum, at $\kappa \gtrsim 25$), whereas the paper shows a well-defined peak of about 430 nA at 100 Hz. Two comparisons are possible:

- **Height and existence of the peak at high κ.** Not reproduced. This is a disagreement between the present model and the published simulation at $\kappa \gg 1$, not a numerical instability of the present one (Part C).
- **Peak width.** The published net peak at 100 Hz, 25 mV has a full width at half maximum of about 80 mV (read from the published figure, ±10 mV). No single rate constant in the present model gives a width above 64 mV (Figure 16B).

The paper notes that a peak-height analysis of the Lovrić type, which is the class of model used here, places the rate constant about two orders of magnitude lower than the full numerical simulation and attributes the difference to background at high frequency [1]. That is not sufficient to explain a peak that the present model puts at less than 10^{-19} of its maximum. The cause has not been established. Candidate explanations, none tested, are: (i) the instrument records a time-averaged rather than end-of-pulse current; (ii) the real MB population has a spread of rate constants (from packing, orientation or tilt), so that a sub-population with $\kappa \approx 0.44$ produces the peak even when the mean rate is very fast; (iii) Butler–Volmer kinetics fail at the largest rates. Explanation (ii) is consistent with the present model's own conclusion in Part D: the signal always comes from labels with $\kappa \approx 0.44$, whatever the mean.

The same mismatch appears for ferrocene. The literature rate constant for ferrocene-hexanethiol (2.7×10^5 s^{-1} [1]) gives κ of order 10^4 at 10 Hz, where the model predicts no signal, yet peaks of 5×10^{-8} to 10^{-5} A were seen on the device [12]. This is one reason ferrocene is not included in the model (Part F).

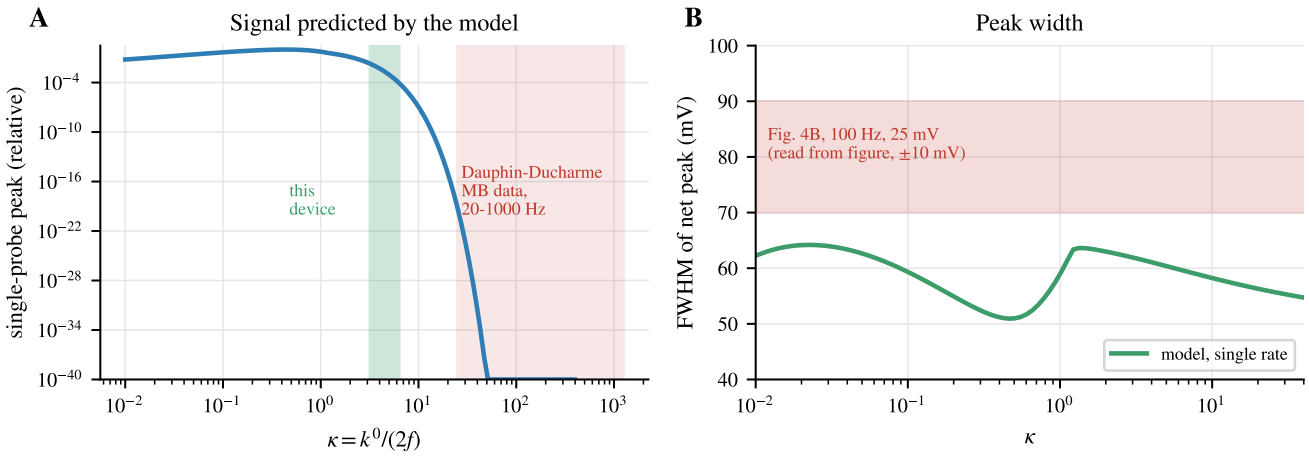


Figure 16: **A:** the single-probe peak signal against κ (curve) with the range of κ corresponding to the Dauphin-Ducharme MB data (red) and to a label at the PEG6 tip of this device (green). **B:** full width at half maximum of the modelled peak against κ compared with the width read from the published Figure 4B (band, ± 10 mV).

What this means for the device. At 10 to 20 Hz the signal comes from labels with $\kappa \approx 0.44$ by construction, and it is largely insensitive to the anchoring rate constant (Part D). If the real rate distribution differs from the one assumed, the *absolute* current would differ, and it would not scale linearly with frequency. The proportionality of the peak to intact probe (Eq. V.2), which is what the readout uses, would not change, since it depends only on the assumption that cleavage is independent of label height (Part F).

Discriminating measurements

Two measurements on the fabricated electrode, each achievable with the existing instrumentation:

1. **Peak height against frequency.** The model predicts a straight line through the origin from 1 to about 100 Hz (a ratio of 2.04 for 20 Hz against 10 Hz; Figure 14C). Departure from this would identify the rate distribution.
2. **Peak width.** The model predicts 58 mV at 10 Hz; a substantially wider peak indicates a spread of environments not represented in the model.

Result

The equations and their implementation are correct; the model agrees with the qualitative behaviour of square-wave voltammetry; it does not reproduce published high-rate voltammograms, and it is unvalidated against the device itself. The absolute predicted current should be read as an estimate, with the uncertainty shown in Part H; the proportional relation between peak height and intact probe is a structural property of the model and not a fitted result.

VII. Part F: Readout of the Cleavage Fraction from the MB Signal

Overview

Objective. To state the scope and assumptions of a readout based on the MB signal alone. **Result.** Ferrocene reacts at a potential far from that of MB and does not overlap the MB scan window. The readout F_k^{rec} of Eq. (I.1) compares each electrode with its own initial scan, so the number of probes cancels. The readout does not provide a check that the signal was unchanged by processes other than cleavage; this is stated as assumption A1.

Fc-hexanethiol on the electrode

- **Ferrocene-hexanethiol is co-immobilised with the probe in the same step** and remains on the electrode; it acts as the diluent [12] *(established).*
- **Its formal potential** has been observed at about +0.30 to +0.40 V against Ag/AgCl, varying between electrodes, compared with a literature value near +0.21 V *(assumed).* A single-component ferrocene scan is required to fix it.
- **No separate pair of MB and ferrocene peaks** has been resolved in the recorded scans [12]; the cause has not been established. One possibility is that the two labels were scanned in separate voltage windows (MB about −0.40 to −0.10 V, ferrocene about +0.15 to +0.55 V), so that one scan would not show both; this has not been checked.
- **Ferrocene's rate constant and coverage are not known.** The literature value for a different construct is too fast for the present model to give any signal (Part E), yet peaks were observed, so no reliable Fc rate constant is available *(assumed).*

Interaction of Fc-hexanethiol with the MB measurement

Overlap of voltages. MB reacts near −0.25 V and the MB scan runs from −0.40 to −0.10 V; ferrocene reacts at +0.21 to +0.40 V, at least 310 mV beyond the end of the MB window (Figure 17A). The shape of a one-electron surface wave, normalised to its peak, is $4u/(1+u)^2$ with $u = e^{\xi}$ (the shape of Part C with $n_e = 1$, scaled by 4 so that the peak is 1). Far from the peak, $|\xi| \gg 1$: for $\xi \ll -1$, $u \to 0$ and $4u/(1+u)^2 \to 4u = 4e^{-|\xi|}$; for $\xi \gg 1$, $(1+u)^2 \to u^2$ and the wave tends to $4/u = 4e^{-|\xi|}$. Either way the wave falls as $4e^{-|\xi|}$. At a distance $\Delta E = 310$ mV, $|\xi| = F\Delta E/RT = 12.1$ and the wave is 2×10^{-5} of the ferrocene peak (Figure 17B). The largest ferrocene peaks recorded, 10^{-5} A, are 627 times the predicted MB peak, so in the worst case the ferrocene tail would add 1.44% to the MB peak. The Butler–Volmer model of Part C predicts a far smaller tail (below 10^{-100}). Interference by the ferrocene wave is therefore negligible or, in the worst case, small.

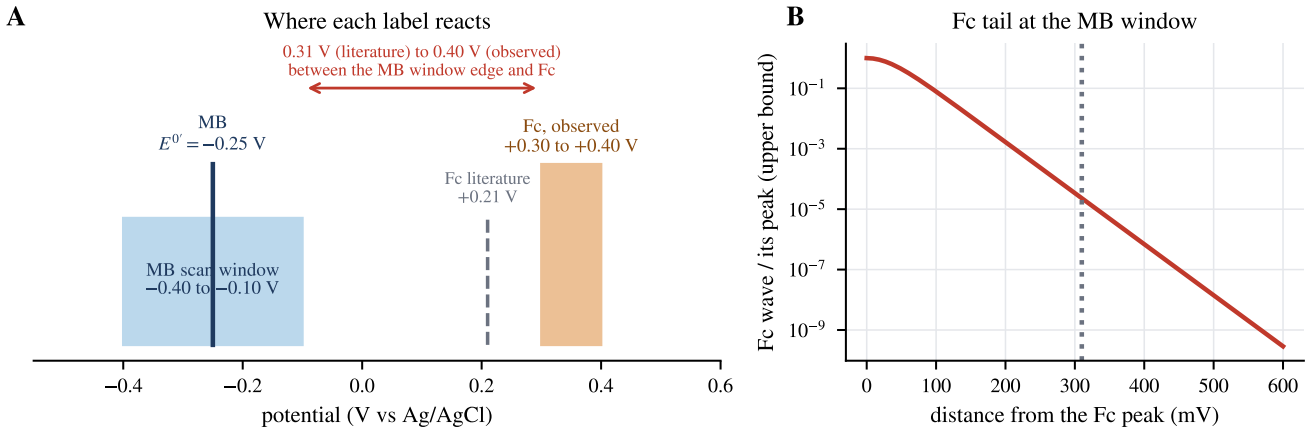


Figure 17: **A:** the voltage of the MB scan window and formal potential, and the two estimates for ferrocene. **B:** upper bound on the ferrocene wave (relative to its peak) at a given distance from its peak; the dotted line is the distance from the ferrocene literature value to the edge of the MB window.

Physical effect on MB. Ferrocene-hexanethiol chains are neighbours of the probe. They may crowd the tether and change $P(d)$, and their charge state could shift $E^{0\prime}$ of MB. Neither has been measured; the first is the reason d_{PEG} is treated as uncertain (Part H).

Comparison with a ferrocene-referenced readout

A readout referenced to the co-immobilised ferrocene would divide the MB peak by the ferrocene peak on the same scan and compare the ratio before and after:

$$(F_k^{\mathrm{rec}})_{\text{with Fc}} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{Fc}}(T)}{\Delta I_{\mathrm{MB}}(0)/\Delta I_{\mathrm{Fc}}(0)} = 1 - \frac{\Delta I_{\mathrm{MB}}(T)}{\Delta I_{\mathrm{MB}}(0)} \quad \text{whenever } \Delta I_{\mathrm{Fc}}(T) = \Delta I_{\mathrm{Fc}}(0), \tag{VII.1}$$

because $\dfrac{\Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{Fc}}(T)}{\Delta I_{\mathrm{MB}}(0)/\Delta I_{\mathrm{Fc}}(0)} = \dfrac{\Delta I_{\mathrm{MB}}(T)}{\Delta I_{\mathrm{MB}}(0)} \cdot \dfrac{\Delta I_{\mathrm{Fc}}(0)}{\Delta I_{\mathrm{Fc}}(T)}$, and the last factor equals 1 when the ferrocene signal is unchanged, so the ferrocene terms cancel exactly. Eq. (VII.1) equals Eq. (I.1) whenever the ferrocene signal is unchanged by incubation. A ferrocene reference therefore does not correct for differences in coverage between electrodes; Eq. (I.1) already does that by comparing each electrode only with itself. What the ferrocene reference could add is a *diagnostic*: if $\Delta I_{\mathrm{Fc}}(T) \neq \Delta I_{\mathrm{Fc}}(0)$, something other than cleavage changed the electrode. The model does not include that, and neither does the readout. Because ferrocene remains on the electrode this diagnostic remains available experimentally without any change to the model.

Assumptions of the readout

- **A1. No non-enzymatic loss of signal during incubation** *(assumed).* If a fraction δ of the probe is lost by other means (desorption, fouling, electrode ageing), then the intact fraction is $(1 - F_k)(1 - \delta)$ and

$$F_k^{\mathrm{rec}} = 1 - (1 - F_k)(1 - \delta) = F_k + \delta\,(1 - F_k). \tag{VII.2}$$

  Expanding, $F_k^{\mathrm{rec}} = 1 - (1 - F_k) + \delta(1 - F_k) = F_k + \delta(1 - F_k)$. The bias $\delta(1 - F_k)$ is always positive: drift is read as extra cleavage, so a chronic wound would read as more severe than it is (Figure 18). In a sample with no MMP-9 ($F_k = 0$) the readout is $F_k^{\mathrm{rec}} = \delta$, and solving Eq. (VIII.2) for the concentration that would give this cleavage fraction, $c = -\ln(1 - \delta)/(k_{\mathrm{eff}}T)$, a loss of 5% at 30 min would read as 2.1 nM, about one third of the healing/chronic boundary.

- **A2. Cleavage is equally likely for any label height** d *(assumed).* Suppose instead the relative cleavage rate is $q(d) \propto e^{\lambda(d-\bar{d})}$, normalised to the same mean, so $\lambda > 0$ means labels far from the gold are cut faster (as steric hindrance

near the surface would produce). Explicitly, $q(d) = e^{\lambda(d-\bar{d})}/\int P\, e^{\lambda(d-\bar{d})}\mathrm{d}d$, a probe at height d survives an incubation with probability $s(d) = e^{-\bar{k}\,q(d)}$ (with $\bar{k} = k_{\text{eff}}[\text{MMP-9}]T$ the mean exponent), the true cleavage fraction is $1 - \int P\, s\,\mathrm{d}d$, and the recovered fraction is $1 - \Delta I_{\text{pk}}[P\,s]/\Delta I_{\text{pk}}[P]$, evaluated with Eq. (V.1). The surviving population then over-represents near labels, which dominate the signal, and F_k^{rec} falls below the true fraction cut. For $\lambda = 2$ nm^{-1} and a true fraction of 27% the recovered fraction is 18 points too low (Figure 19). If the sign of λ is reversed the error is positive.

- **A3. The two scans are made on the same electrode with the same settings.** Different frequency, amplitude or window would change the peak independent of cleavage.
- **A4. Noise in each reading is independent**, as used below.

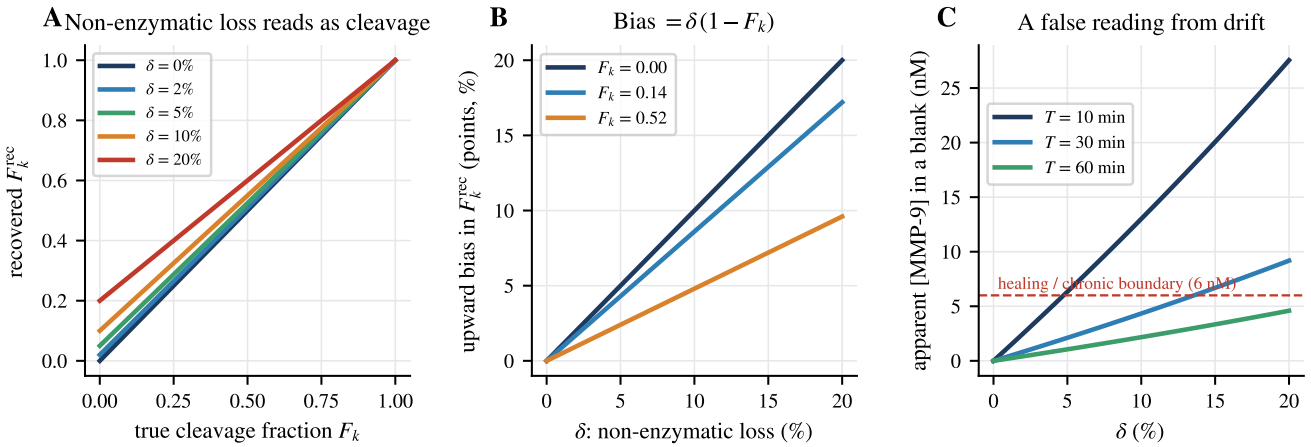


Figure 18: Assumption A1. **A:** recovered against true cleavage fraction for several levels of non-enzymatic loss δ. **B:** the resulting upward bias in points of F_k, $\delta(1 - F_k)$. **C:** the MMP-9 concentration that a blank sample (none present) would appear to contain, using the kinetics of Part G with $\eta = 0.20$; the dashed line is the healing/chronic boundary.

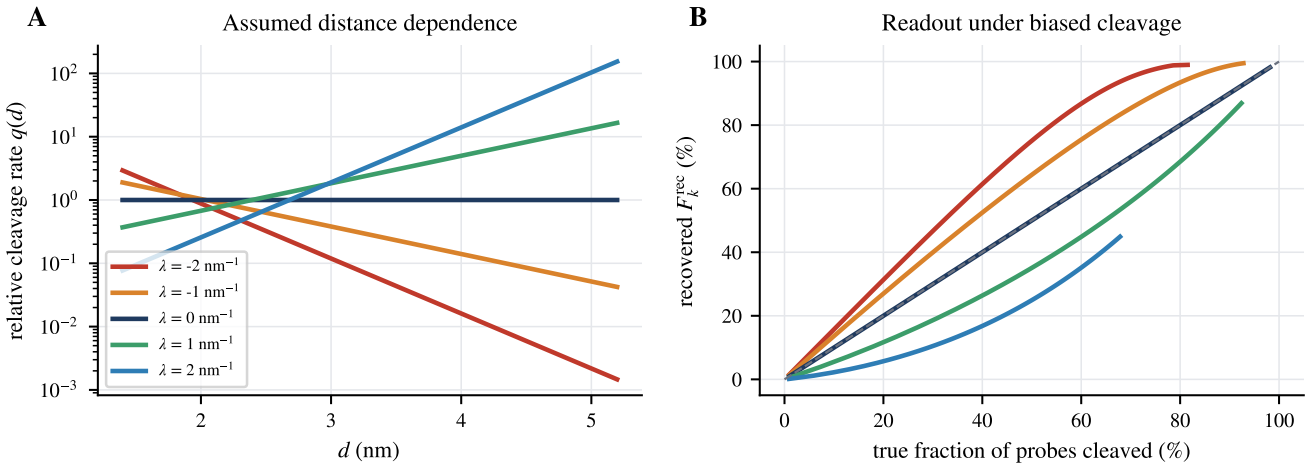


Figure 19: Assumption A2. **A:** relative cleavage rate against label height for several values of λ (all with the same average). **B:** recovered cleavage fraction against the true fraction of probes cut; the dashed diagonal is the case of no bias ($\lambda = 0$). If labels near the gold are cut more slowly, the readout underestimates.

Noise, coverage and electrode-to-electrode variation

Noise in the readout. Let each peak height have an independent absolute error σ_I, and $\sigma_r = \sigma_I/\Delta I_{\text{MB}}(0)$. For $F_k^{\text{rec}} = 1 - I_T/I_0$ the partial derivatives are $\partial F_k^{\text{rec}}/\partial I_T = -1/I_0$ and $\partial F_k^{\text{rec}}/\partial I_0 = I_T/I_0^2$. For independent errors the variances add, each weighted by the square of its derivative:

$$\sigma_F^2 = \left(\frac{1}{I_0}\right)^2\sigma_I^2 + \left(\frac{I_T}{I_0^2}\right)^2\sigma_I^2 = \frac{\sigma_I^2}{I_0^2}\left[1 + \left(\frac{I_T}{I_0}\right)^2\right] = \sigma_r^2\left[1 + (1 - F_k)^2\right], \qquad \sigma_F \xrightarrow{F_k \to 0} \sqrt{2}\,\sigma_r. \tag{VII.3}$$

The noise floor of the before/after MB readout has not been measured *(assumed)*. A reference case of $\sigma_r = 2.4\%$ per reading is used in Part G; by Eq. (VII.3) at $F_k \to 0$ it gives $\sigma_F = \sqrt{2} \times 2.4 = 3.4\%$.

Coverage. By Eq. (V.2), $\Delta I(E; F_k) = (1 - F_k)\,\Gamma_{\text{tot}}\,C(E)$ with $C(E)$ independent of Γ_{tot}, so $F_k^{\text{rec}} = 1 - (1 - F_k)\Gamma_{\text{tot}}C/(\Gamma_{\text{tot}}C) = F_k$. Hence F_k^{rec} is independent of Γ_{tot} exactly (three electrodes with a tenfold spread in coverage give the same F_k^{rec} to machine precision, Figure 20A). Coverage matters only through noise: with a fixed absolute noise, an electrode with less probe has a smaller I_0 and so a larger σ_r, worsening the detection limit (Figure 20C). Electrode-to-electrode variation therefore does not affect a calibration of F_k^{rec} against concentration, which does not use the absolute current; the remaining effects are (a) the noise penalty on low-coverage electrodes, and (b) variation between electrodes in the surface efficiency η of Part G, which is unknown *(assumed)*.

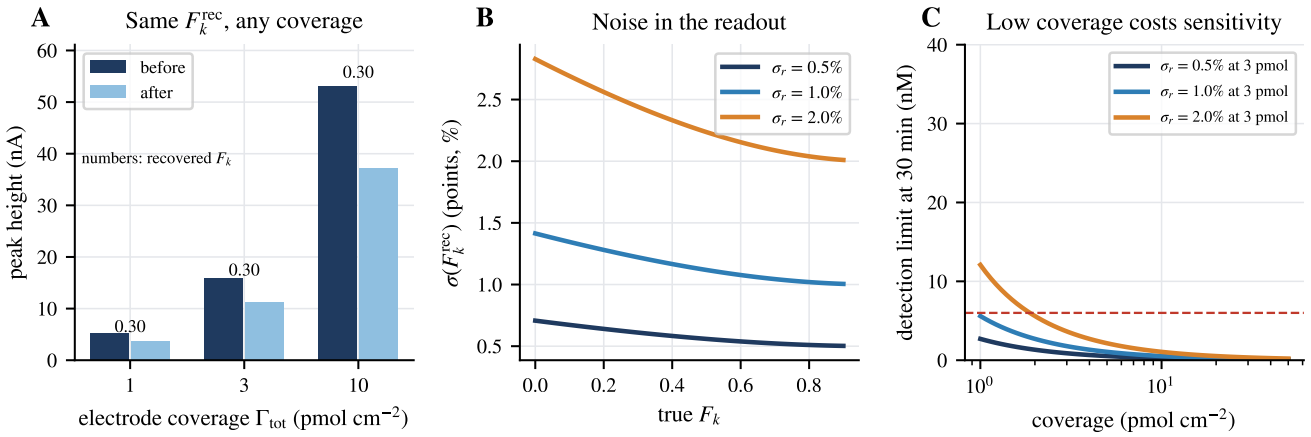


Figure 20: **A:** peak heights before and after 30% cleavage on three electrodes of different coverage; the recovered fraction (numbers) is identical. **B:** noise of the readout from Eq. (VII.3) for three per-reading noise levels. **C:** detection limit at 30 min against coverage for a fixed absolute noise (levels shown are the noise at 3 pmol cm^{-2}); low coverage degrades it.

Result

The MB scan and the algebra of the readout are unaffected by the Fc-hexanethiol diluent. The readout rests on assumptions A1 to A4; A1 and A2 are testable, and a ferrocene signal recorded on the same electrode would test A1. Coverage variation affects the noise, not the readout. The noise floor is unknown and controls the detection limit (Part G).

VIII. Part G: From MMP-9 Concentration to Predicted Signal

Overview

Objective. To relate the MMP-9 concentration and incubation time to the cleavage fraction and to determine the detection limit. **Result.** Each intact probe is cleaved at a rate proportional to the MMP-9 concentration, so the intact fraction decays exponentially, $F_k = 1 - e^{-k_{\text{eff}}[\text{MMP-9}]T}$. With the measured solution rate constant and a surface efficiency of 0.20 (assumed), 6 nM (healing) gives $F_k = 14\%$ and 30 nM (chronic) $F_k = 52\%$ after 30 minutes. The detection limit is set by the readout noise, which has not been measured.

Assumptions

- **The solution rate constant applies to the tethered probe, scaled by a surface efficiency η.** The measured value, $k_{\text{cat}}/K_m = 67500$ M^{-1}s^{-1} ($k_{\text{cat}} = 86.4$ s^{-1}, $K_m = 1.3$ mM), is for the free peptide SGKIPRTLTA in solution [7] *(established)*. On the surface the enzyme must reach the probe and the tether is attached to a PEG spacer, a dye and a crowded layer, so cleavage is expected to be slower. $\eta = 0.20$ is a central estimate, with a plausible range of 0.01 to 1, not measured for this device *(assumed)*. It would be measured by comparing the electrode readout with a solution-phase fluorescence assay on the same sample.
- **The enzyme concentration is constant during the incubation** (enzyme is a catalyst and is not consumed; loss by adsorption to surfaces or by inhibitors is not included). [MMP-9] means active enzyme; activation of the pro-form and inhibition by TIMPs are treated in the wound-biology report.
- **The conditions for first-order kinetics below hold.** They are checked explicitly.

Equations and derivation

Cleavage rate of a fixed site. For an enzyme E cutting a substrate S that is fixed on a surface,

$$\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{ES} \xrightarrow{k_{\mathrm{cat}}} \mathrm{E} + \text{products}, \qquad K_m = \frac{k_{-1} + k_{\mathrm{cat}}}{k_1}.$$

Let S_T be the number of intact probes (free plus enzyme-bound). If the enzyme-bound state reaches steady state quickly compared with the depletion of probe, then $d[\mathrm{ES}]/dt \approx 0$, i.e. $k_1[\mathrm{E}]\,[\mathrm{S}] = (k_{-1} + k_{\mathrm{cat}})[\mathrm{ES}]$ with $[\mathrm{S}] = S_T - [\mathrm{ES}]$. Dividing by k_1 and using the definition of K_m: $[\mathrm{E}](S_T - [\mathrm{ES}]) = K_m[\mathrm{ES}]$, so $[\mathrm{E}]S_T = ([\mathrm{E}] + K_m)[\mathrm{ES}]$ and $[\mathrm{ES}] = S_T\,[\mathrm{E}]/(K_m + [\mathrm{E}])$. Each cleavage removes one probe, so

$$\frac{dS_T}{dt} = -k_{\mathrm{cat}}[\mathrm{ES}] = -\frac{k_{\mathrm{cat}}[\mathrm{E}]}{K_m + [\mathrm{E}]}\,S_T \;\approx\; -\frac{k_{\mathrm{cat}}}{K_m}\,[\mathrm{E}]\,S_T \quad \text{for } [\mathrm{E}] \ll K_m. \tag{VIII.1}$$

This is first-order in the probe. The condition is that the *enzyme* concentration is small compared with K_m: 30 nM against $K_m = 1.3$ mM is a ratio $[\mathrm{E}]/K_m = 30 \times 10^{-9}/1.3 \times 10^{-3} = 2.3 \times 10^{-5}$. The relevant condition is not that the enzyme exceeds the probe in number: the enzyme is a catalyst, and what matters is that it is not depleted and does not saturate the probes. Two further requirements are met. The enzyme bound at any time is at most $S_T\,[\mathrm{E}]/K_m \leq$ 51 fmol $\times\, 2.3 \times 10^{-5} = 1.2 \times 10^{-3}$ fmol, negligible against the enzyme in even 1 µL of a 6 nM sample (6 fmol), so free and total enzyme concentrations are equal. And the steady-state approximation requires the enzyme–probe complex to equilibrate faster than the probes are depleted: equilibration takes about $1/(k_{-1} + k_{\mathrm{cat}}) \leq 1/k_{\mathrm{cat}} \approx 12$ ms, against depletion times of minutes to hours (Eq. VIII.2).

Cleavage fraction. Including the surface efficiency, $k_{\mathrm{eff}} = \eta\,(k_{\mathrm{cat}}/K_m)_{\mathrm{sol}}$ (13500 $\mathrm{M^{-1}s^{-1}}$ at $\eta = 0.20$). Eq. (VIII.1) is separable: $dS_T/S_T = -k_{\mathrm{eff}}[\text{MMP-9}]\,dt$; integrating from 0 to T gives $\ln[S_T(T)/S_T(0)] = -k_{\mathrm{eff}}[\text{MMP-9}]T$, so $S_T(T) = S_T(0)\,e^{-k_{\mathrm{eff}}[\text{MMP-9}]T}$, and with $F_k \equiv 1 - S_T(T)/S_T(0)$,

$$F_k(T) = 1 - e^{-k_{\mathrm{eff}}\,[\text{MMP-9}]\,T}, \qquad t_{1/2} = \frac{\ln 2}{k_{\mathrm{eff}}[\text{MMP-9}]} \quad \left(\text{from } F_k = \tfrac{1}{2} :\; e^{-k_{\mathrm{eff}}[\text{MMP-9}]t_{1/2}} = \tfrac{1}{2}\right). \tag{VIII.2}$$

Worked example. At 30 nM and 30 min, $k_{\mathrm{eff}}[\text{MMP-9}]T = 13500 \times 30 \times 10^{-9} \times 1800 = 0.729$, so $F_k = 1 - e^{-0.729} = 52\%$. At 6 nM, 0.146 and 14%. At 6 nM, $k_{\mathrm{eff}}[\text{MMP-9}] = 13500 \times 6 \times 10^{-9} = 8.1 \times 10^{-5}$ $\mathrm{s^{-1}}$, so $t_{1/2} = 0.693/(8.1 \times 10^{-5}) = 8.56 \times 10^3$ s=143 min; at 30 nM, 4.05×10^{-4} $\mathrm{s^{-1}}$ gives $t_{1/2} = 1.71 \times 10^3$ s=29 min.

Results

Wound state	[MMP-9]	F_k at 30 min ($\eta = 0.20$)	Half-time of probe	Source
Healing	6 nM	14%	143 min	Trengove 1999 [7b]
Chronic	30 nM	52%	29 min	Trengove 1999 [7b]

The two states are best separated by the readout at about 83 min, where F_k differs by 53 points, but the difference is already large by 30 min (Figure 21, Figure 22).

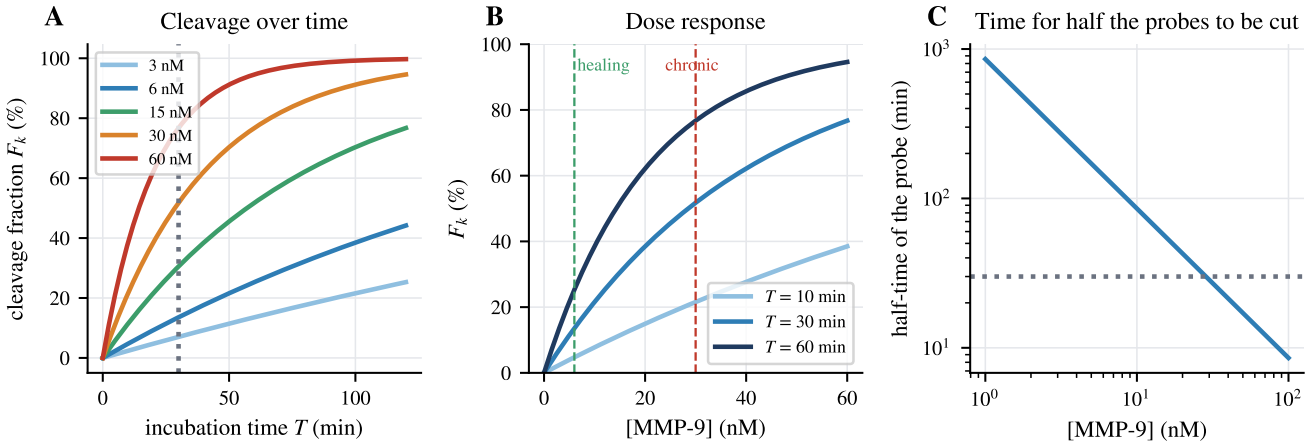


Figure 21: **A:** cleavage fraction against incubation time for five concentrations ($\eta = 0.20$). **B:** cleavage fraction against concentration at three incubation times; healing and chronic concentrations are marked. **C:** the time for half of the probes to be cut.

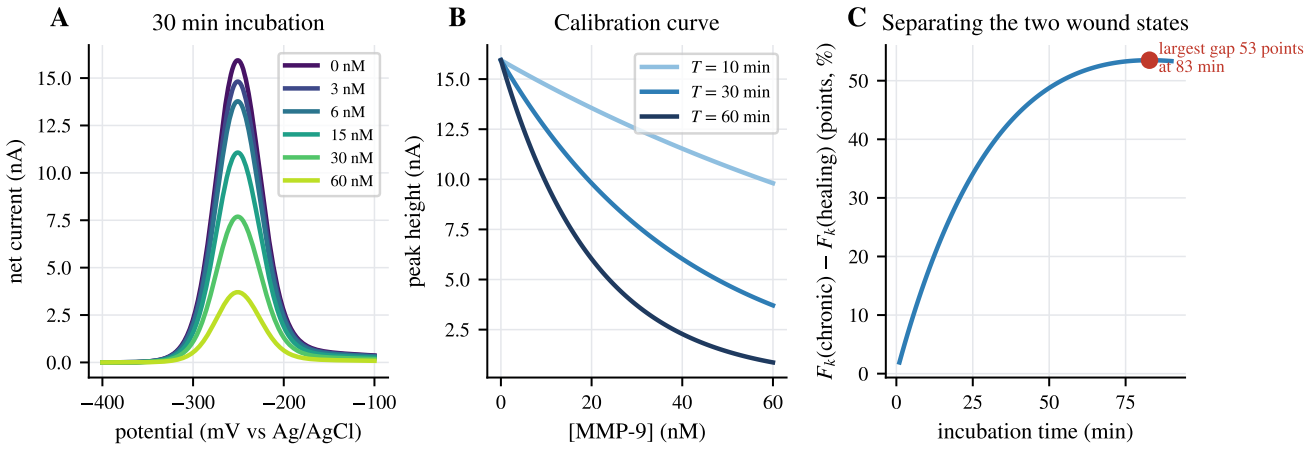


Figure 22: **A:** predicted voltammograms after 30 min for a range of MMP-9 concentrations (the concentration chain of Figure 4 end to end). **B:** predicted peak height against concentration for three incubation times. Absolute values depend on the assumed inputs (Part H). **C:** the difference in cleavage fraction between chronic and healing samples against time.

Detection limit

A concentration is detected when the readout exceeds z times its noise; a criterion of $z = 3$ is used. The smallest detectable cleavage fraction is $F_{\min} = z\,\sigma_F$. Setting $F_{\min} = 1 - e^{-k_{\text{eff}}cT}$ in Eq. (VIII.2) gives $e^{-k_{\text{eff}}cT} = 1 - F_{\min}$, and taking logarithms, $-k_{\text{eff}}cT = \ln(1 - F_{\min})$, gives the lowest detectable concentration c:

$$c_{\text{LOD}} = \frac{-\ln\!\left(1 - z\,\sigma_F\right)}{k_{\text{eff}}\,T}. \tag{VIII.3}$$

Worked example. A reference case $\sigma_F = 3.4\%$ ($\sigma_r = 2.4\%$ per reading) gives $F_{\min} = 3 \times 0.034 = 0.102$ and, at 30 min ($T = 1800$ s), $c_{\text{LOD}} = -\ln(1 - 0.102)/(13500 \times 1800) = 0.1076/(2.43 \times 10^7) = 4.4 \times 10^{-9}$ M= 4.4 nM, below the 6 nM boundary. The readout noise has not been measured *(assumed)*. In general, detection of the healing concentration (6 nM) in 30 min requires $\sigma_F < 4.5\%$, i.e. $\sigma_r < 3.2\%$ per reading (Eq. VII.3). The full dependence on noise, time and η is shown in Figure 23.

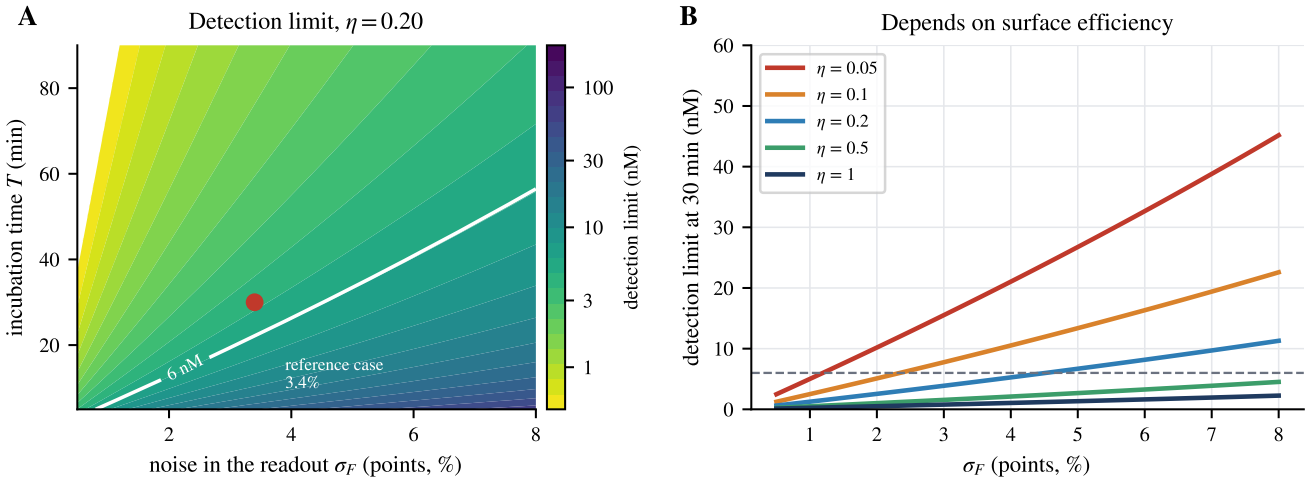


Figure 23: **A:** detection limit (colour, log scale) against readout noise and incubation time for $\eta = 0.20$; the white contour is the 6 nM healing/chronic boundary. The red dot marks the reference case $\sigma_F = 3.4\%$. **B:** detection limit against noise at 30 min for five surface efficiencies.

Sensitivity to the surface efficiency

Since η is unmeasured, its influence was examined across a log-uniform range 0.01 to 1 (Figure 24). For η below 0.15 the healing concentration falls below the detection threshold at 30 min (for $\sigma_F = 3.4\%$); over the log-uniform range, 41% of the values of η give a 6 nM readout above threshold at 30 min. At the central value, about 22 min of

incubation is needed to detect 6 nM. Incubation time is the easiest lever for a low η. Because η multiplies the rate, an empirical calibration of F_k^{rec} against known MMP-9 concentrations on the fabricated electrode replaces the assumed η directly, and is preferable to using the model's value.

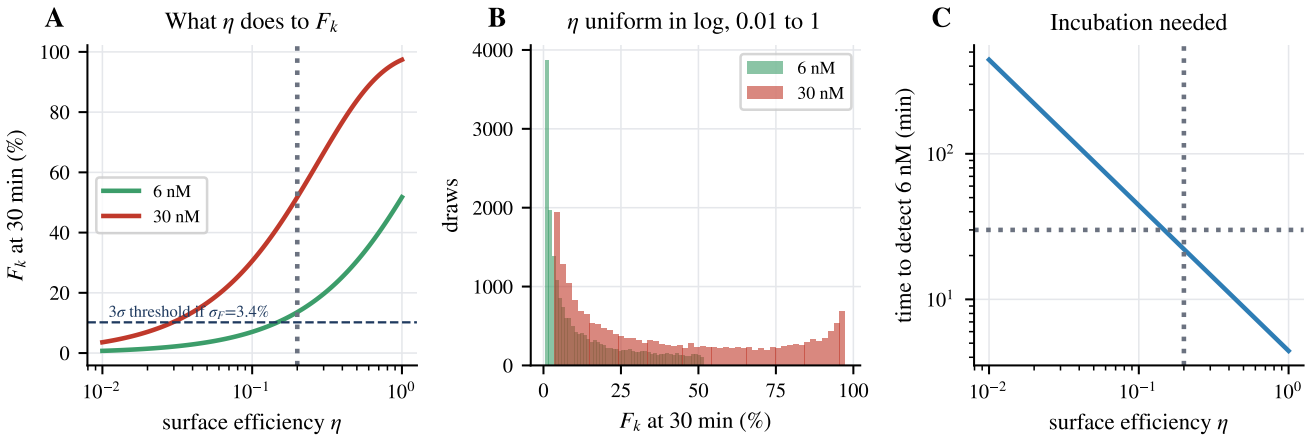


Figure 24: **A:** cleavage fraction at 30 min against η for healing and chronic concentrations; the dashed line is the detection threshold if $\sigma_F = 3.4\%$. **B:** distribution of the cleavage fraction when η is drawn log-uniformly from 0.01 to 1. **C:** incubation time needed to detect 6 nM against η.

Result

Eq. (VIII.2) converts concentration and time to cleavage fraction, with the solution rate constant confirmed and a surface efficiency that is not. Whether the device can resolve healing from chronic samples in about 30 minutes depends on two numbers not yet known: η and the noise of the before/after readout.

IX. Part H: Sensitivity to the Assumed Inputs

Overview

Objective. To determine which assumed inputs affect which predictions. **Result.** The absolute current depends strongly on the tether geometry and coverage and is uncertain by a factor of several. The cleavage readout is independent of these inputs; it depends on assumptions A1 and A2 (Part F), the noise, and the surface efficiency η.

Method

Each input was varied over a stated range with the others held at their base values, and the predicted peak height at 10 Hz recomputed (Figure 25A and the table below). The ranges are: Γ_{tot} a factor of three either way; area ±5%; frequency 10 to 20 Hz (the confirmed range); d_{PEG} from the ideal-chain to the fully stretched value; β 5 to 13 nm^{-1} with the measured rate at 0.8 nm held fixed; k_{ref} a factor of 10 either way; α 0.3 to 0.5; and the stricter wall model of Part B.

Parameter varied	low	high
coverage Γ_{tot} (÷3, ×3)	−67%	+200%
frequency (10 to 20 Hz)	+0%	+104%
tip position d_{PEG} (1.26 to 2.1 nm)	−5%	−94%
tether model (no bead below tip)	−38%	+0%
decay constant β (5 to 13 nm^{-1})	−4%	−23%
rate anchor k_{ref} (×0.1, ×10)	−15%	−10%
area A (±5%)	−5%	+5%
transfer coefficient α (0.3 to 0.5)	−1%	−0%

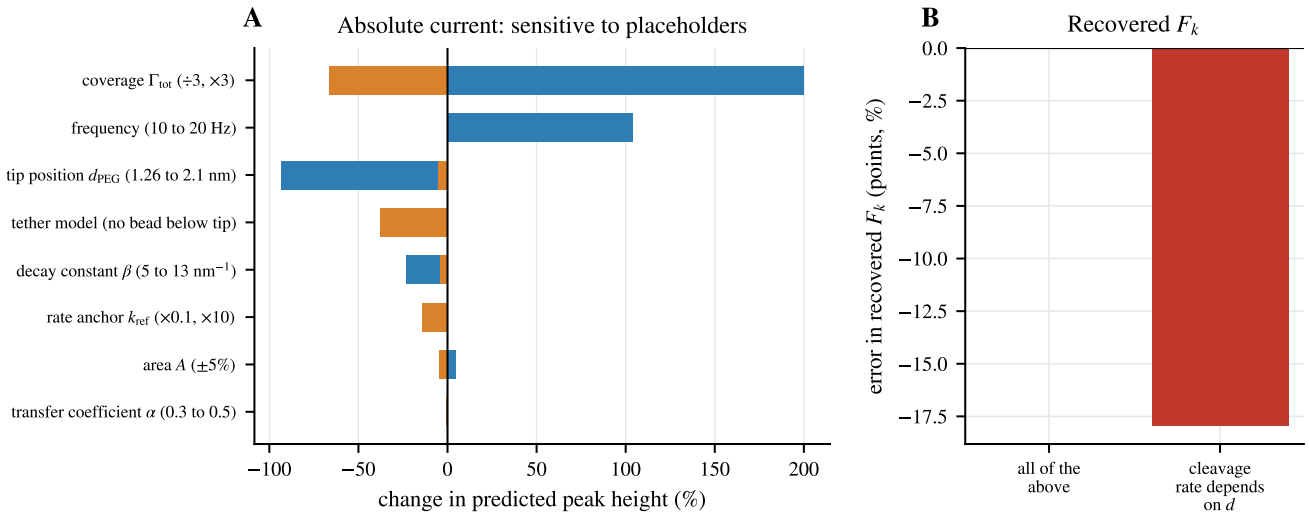


Figure 25: **A:** change in the predicted peak height when each input is varied over its stated range (blue: high end, orange: low end). **B:** the error in the recovered cleavage fraction caused by a distance-dependent cleavage rate (Assumption A2, $\lambda = 2$ nm^{-1}); the variations in A produce none, because the readout is a ratio of two scans of the same electrode.

Results

- **Coverage and frequency** scale the current directly: a factor of three in Γ_{tot} or a doubling of frequency changes the peak by the same factor.
- **The height of the PEG6 tip** is the most sensitive geometric input, because the rate constant falls tenfold per 0.23 nm: a tip 0.7 nm higher than assumed removes most of the signal, while a lower one changes it little. The tunnelling decay constant and the choice of tether model matter at the tens-of-percent level. The anchoring rate k_{ref} and the transfer coefficient α have little effect (Part D).
- **None of these** changes F_k^{rec}: they multiply the current before and after cleavage alike.

Input	Absolute current	Readout F_k^{rec}	Estimated concentration
Γ_{tot}, A, f, d_{PEG}, β, k_{ref}, α	strong	none	none (through F_k)
Non-enzymatic loss δ (A1)	none	upward bias	overestimate
Distance-dependent cleavage (A2)	none	bias either way	bias either way
Surface efficiency η	none	none	scales with $1/\eta$
Readout noise σ_r	none	random error	sets the detection limit
Coverage (via noise)	none	random error	detection limit

Result

The quantities that most affect the device's usefulness are η, the noise, and the assumptions A1 and A2, and none of them is constrained by the assumed inputs that dominate the predicted current. The current the model predicts (15.9 nA at 10 Hz) is therefore an estimate with an uncertainty of a large factor, and it should not be quoted as a specification; the ratio-based readout and its logic are much better founded.

X. Summary of Results, Assumptions and Open Items

Part	Component	Status	
A	One-pot protocol, Fc-hexanethiol diluent, no MCH	Current protocol [12]	*(established)*
A	Electrode area (1.7 mm^2)	Confirmed against the fabricated device	*(established)*
A	Probe coverage Γ_{tot}	Engineering estimate; not measurable with available instrumentation	*(assumed)*
A	Surface composition (probe : Fc-hexanethiol)	Not the solution ratio; unmeasured	*(assumed)*
B	Freely jointed chain, half-Gaussian $P(d)$	Derived and checked against exact chains; not validated on the real surface	*(assumed)*
B	PEG6 tip height, tunnelling decay constant β	Estimates (Flory; alkanethiol value)	*(assumed)*
C	Butler–Volmer ODE, closed-form solution, stable current	Exact mathematics, verified numerically	*(established)*
C	SWV amplitude (25 mV), frequency (10 to 20 Hz)	Confirmed against potentiostat settings	*(established)*
D	Peak proportional to intact probe	Exact given A2	*(established)*
D	Signal band ∼0.27 nm above the tip at 10 Hz	Prediction of the model	*(assumed)*
E	Reproduction of published high-rate voltammograms	Not reproduced; cause unknown	*(assumed)*
E	Comparison with the fabricated device	Not yet done	*(assumed)*
F	Fc-hexanethiol diluent	Outside the scope of the model; no interference with the MB window	*(established)*
F	No non-enzymatic loss (A1), cleavage independent of label height (A2)	Assumed; both testable	*(assumed)*
F	Noise floor of the before/after readout	Not measured; reference case 3.4% used	*(assumed)*
G	Solution cleavage rate (Kridel)	Measured for the free peptide	*(established)*
G	Surface efficiency η	Central estimate 0.20, range 0.01 to 1	*(assumed)*

Conclusions. The electrochemistry rests on standard Butler–Volmer theory and an exact solution. Its central structural result, that the peak height is proportional to the number of intact probes, and hence that $F_k^{\mathrm{rec}} = 1 - \Delta I_{\mathrm{MB}}(T)/\Delta I_{\mathrm{MB}}(0)$ is independent of coverage and of most of the uncertain geometry, does not depend on the assumed values. What does depend on them is the absolute current, which is uncertain by a large factor and which is not reproduced by the model against the one published data set at high rate.

Recommended measurements.

1. Peak height and width against frequency on a fabricated electrode (Part E): tests the rate distribution and the absolute-current prediction.
2. A single-component ferrocene scan and a scan of both labels together: fixes the ferrocene potential and shows whether the two are resolved (Part F).
3. A blank sample (no MMP-9) incubated for the full time, with and without a ferrocene check: bounds non-enzymatic loss δ (A1).
4. A calibration of F_k^{rec} against known MMP-9 concentrations, with a parallel solution assay: replaces the assumed η and A2 (Part G).
5. Repeat scans of one electrode: gives the readout noise σ_r directly and hence the detection limit.
6. Surface coverage by an independent method, when available: fixes Γ_{tot}.

Appendix A: Standard Integrals

The integrals used in Parts A and B are derived here from a single result, the Gaussian integral.

Gaussian integral. For $a > 0$ let $I(a) = \int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x$. Its square is a double integral over the plane,

$$I^2 = \int_{-\infty}^{\infty}\!\int_{-\infty}^{\infty} e^{-a(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y.$$

In polar coordinates $x^2 + y^2 = \rho^2$ and the area element is $\rho\,\mathrm{d}\rho\,\mathrm{d}\phi$, so

$$I^2 = \int_0^{2\pi}\mathrm{d}\phi \int_0^{\infty} e^{-a\rho^2}\rho\,\mathrm{d}\rho = 2\pi\left[-\frac{e^{-a\rho^2}}{2a}\right]_0^{\infty} = \frac{\pi}{a},$$

so $I = \sqrt{\pi/a}$, that is

$$\int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}, \qquad \int_0^{\infty} e^{-ax^2}\,\mathrm{d}x = \frac{1}{2}\sqrt{\frac{\pi}{a}}, \tag{X.1}$$

the second form following because the integrand is even.

Normalisation of the normal density, Eq. (III.5)**.** With $a = 1/(2\sigma_z^2)$, $\int_{-\infty}^{\infty} e^{-z^2/2\sigma_z^2}\mathrm{d}z = \sqrt{2\pi\sigma_z^2} = \sigma_z\sqrt{2\pi}$, so

$$\int_{-\infty}^{\infty} f\,\mathrm{d}z = \frac{1}{\sigma_z\sqrt{2\pi}}\cdot\sigma_z\sqrt{2\pi} = 1. \tag{X.2}$$

Second moment of a Gaussian. Differentiating $\int_0^{\infty} e^{-ar^2}\mathrm{d}r = \frac{1}{2}\sqrt{\pi}\,a^{-1/2}$ with respect to a, and noting that $\partial_a e^{-ar^2} = -r^2 e^{-ar^2}$,

$$\int_0^{\infty} r^2 e^{-ar^2}\,\mathrm{d}r = -\frac{d}{da}\left(\frac{\sqrt{\pi}}{2}a^{-1/2}\right) = \frac{\sqrt{\pi}}{4}\,a^{-3/2} = \frac{\sqrt{\pi}}{4a^{3/2}}. \tag{X.3}$$

For the nearest-neighbour distance of Part A, with $a = \pi\rho$: $2\pi\rho\cdot\sqrt{\pi}/(4(\pi\rho)^{3/2}) = 2\pi\rho\sqrt{\pi}/(4\pi^{3/2}\rho^{3/2}) = 1/(2\sqrt{\rho})$.

First moment of a half-Gaussian. With the substitution $w = u^2/(2\sigma_z^2)$, $\mathrm{d}w = u\,\mathrm{d}u/\sigma_z^2$, the limits 0 and ∞ are unchanged and

$$\int_0^{\infty} u\,e^{-u^2/2\sigma_z^2}\,\mathrm{d}u = \sigma_z^2\int_0^{\infty} e^{-w}\,\mathrm{d}w = \sigma_z^2. \tag{X.4}$$

Tail probability of the half-Gaussian. The probability that $u = d - d_{\mathrm{PEG}}$ exceeds U is $\int_U^{\infty} 2f(u)\,\mathrm{d}u = \frac{2}{\sigma_z\sqrt{2\pi}}\int_U^{\infty} e^{-u^2/2\sigma_z^2}\mathrm{d}u$. Substituting $t = u/(\sqrt{2}\sigma_z)$, $\mathrm{d}u = \sqrt{2}\sigma_z\,\mathrm{d}t$, gives $\frac{2}{\sqrt{\pi}}\int_{U/(\sqrt{2}\sigma_z)}^{\infty} e^{-t^2}\mathrm{d}t$. By Eq. (X.1), $\frac{2}{\sqrt{\pi}}\int_0^{\infty} e^{-t^2}\mathrm{d}t = 1$, so with the error function defined as $\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\mathrm{d}t$,

$$\Pr(u > U) = 1 - \mathrm{erf}\!\left(\frac{U}{\sqrt{2}\,\sigma_z}\right). \tag{X.5}$$

With $U = d_{\max} - d_{\mathrm{PEG}} = 3.8$ nm and $\sigma_z = 0.98$ nm the argument is 2.74 and $\Pr = 1.1\times10^{-4}$.

Nomenclature

Term	Meaning	Introduced
MMP-9	the enzyme whose activity the device measures; higher in chronic wounds	Section I
Probe	the peptide SGKIPRTLTA with PEG6 spacer and MB label, attached to the gold	Section I
MB	methylene blue, the redox label that exchanges electrons with the gold	Section I
Fc	ferrocene, the redox group of the diluent Fc-hexanethiol	Section I
Redox	involving gain or loss of electrons; the reduced form has gained them, the oxidised form has lost them	Section I
Voltammogram	graph of current against applied voltage	Section I
SWV	square-wave voltammetry: staircase voltage with a square wave, current sampled at the end of each pulse	Section I
$E^{0\prime}$	formal potential: the voltage at which reduced and oxidised forms are equally favoured (MB: -0.25 V vs Ag/AgCl)	Section I
E_{sw}, f, τ	square-wave amplitude (25 mV), frequency (10 to 20 Hz), and half-period $1/(2f)$	Part C
ΔI	net current, forward minus reverse	Part C
F_k	cleavage fraction: fraction of probes cut	Section I
F_k^{rec}	cleavage fraction recovered from the two peak heights	Section I
Γ_{tot}	probe coverage, $\mathrm{mol\,cm^{-2}}$	Part A
Γ_{red}, Γ_{ox}, Γ_{eq}	amounts of reduced and oxidised MB per area, and the reduced amount at equilibrium	Part C
d, $P(d)$	distance of the MB label from the gold; its probability density	Part B
d_{PEG}	height of the PEG6 tip above the gold	Part B
k^0, k_{ref}	rate constant for electron exchange at $E^{0\prime}$, and its value at the PEG6 tip	Part B
β	tunnelling decay constant: how fast k^0 falls with distance	Part B
k_f, k_b, K	rates of reduction, oxidation, and their sum	Part C
α, n_e	transfer coefficient (0.37) and electrons transferred (2)	Part C
κ	$k^0/(2f)$: label rate relative to pulse rate; best signal at $\kappa \approx 0.44$	Part C
k_{eff}, η	rate constant on the surface and the surface efficiency $k_{\mathrm{eff}}/(k_{\mathrm{cat}}/K_m)_{\mathrm{sol}}$	Part G
δ	fraction of probe lost by means other than cleavage	Part F
σ_r, σ_F	noise in one peak-height reading, relative to the initial peak, and in F_k^{rec}	Part F
FWHM	full width of a peak at half its height	Section I
FJC	freely jointed chain model of a flexible polymer	Part B
Kuhn length	length over which a polymer chain effectively forgets its direction	Part B

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